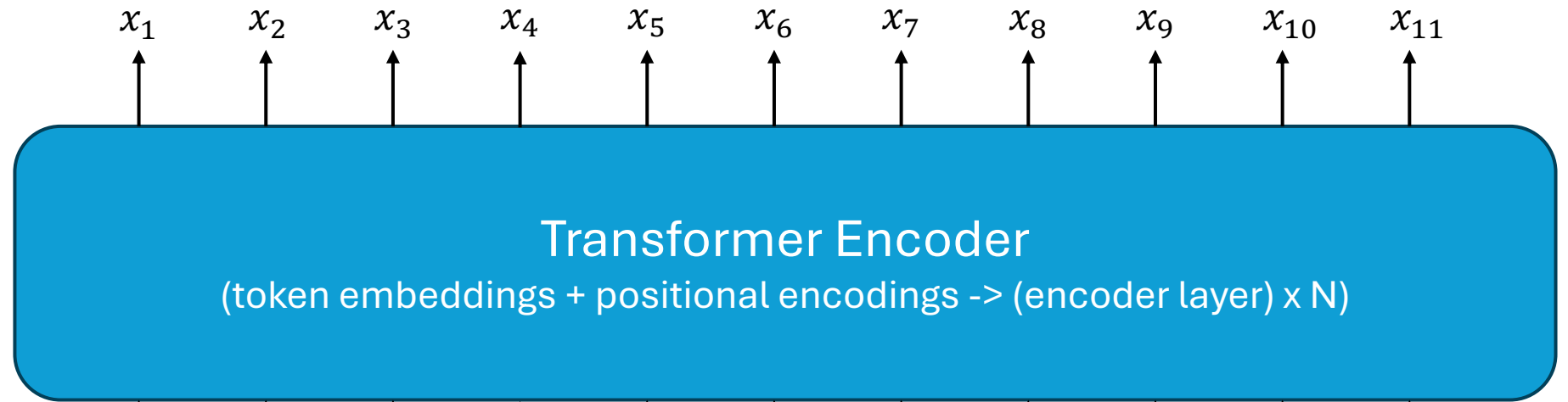


Multimodal data fusion

Matt Engelhard

Recall: training and use of Transformer architectures for text

Output
representations



Input tokens

Even the smallest person can change the course of the future

Today

- Transformers for EHR
- Multimodal data fusion
 - Why?
 - How?
 - Late fusion
 - Joint fusion
 - Early fusion
 - Examples

Moving to the EHR

A foundation model for EHR data

Models for structured EHR data: the “language” of the EHR

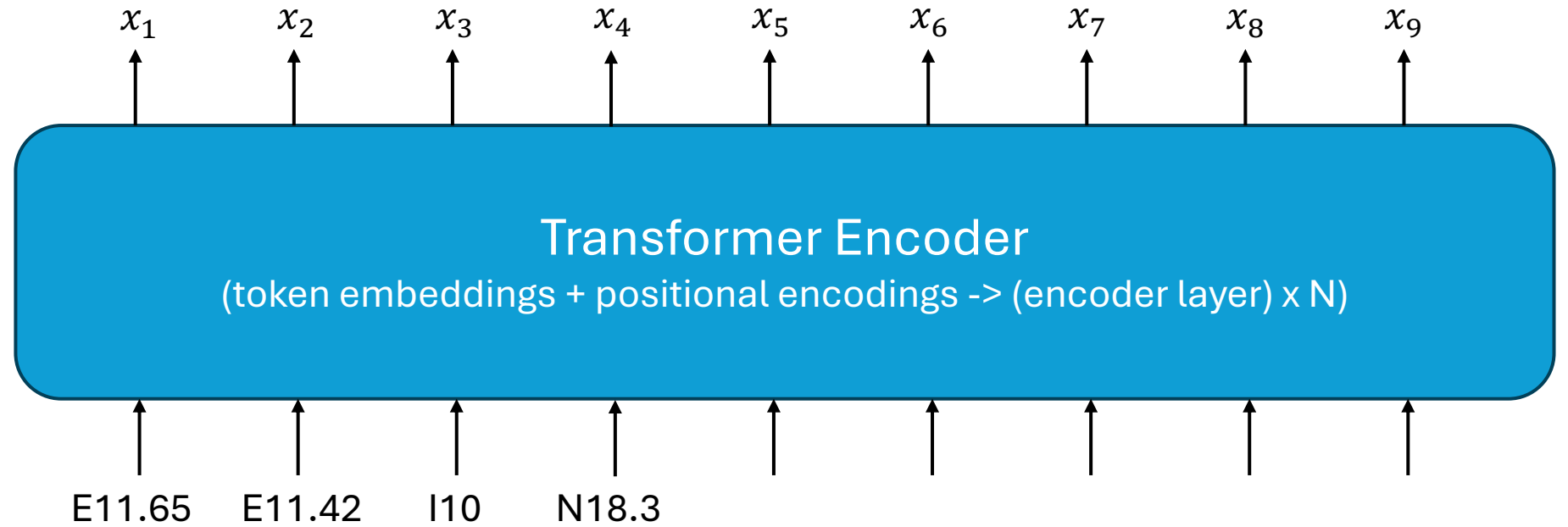
A story about the progression of diabetes...

(“E11.65”, “E11.42”, “I10”, “N18.3”)

- **E11.65** — Type 2 diabetes mellitus with hyperglycemia
- **E11.42** — Type 2 diabetes mellitus with diabetic polyneuropathy
- **I10** — Essential (primary) hypertension
- **N18.3** — Chronic kidney disease, stage 3

Transformers for Electronic Health Records

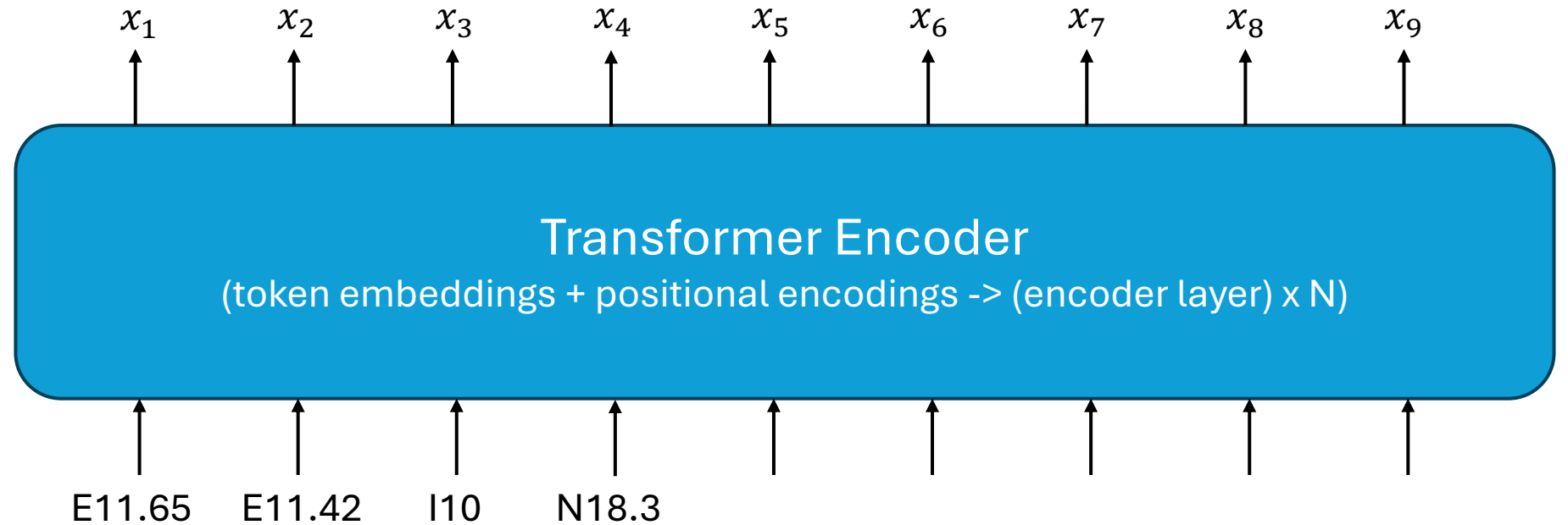
Output
representations



Transformers for Electronic Health Records

Unique consideration 1: we have **position**, **age**, and **time**

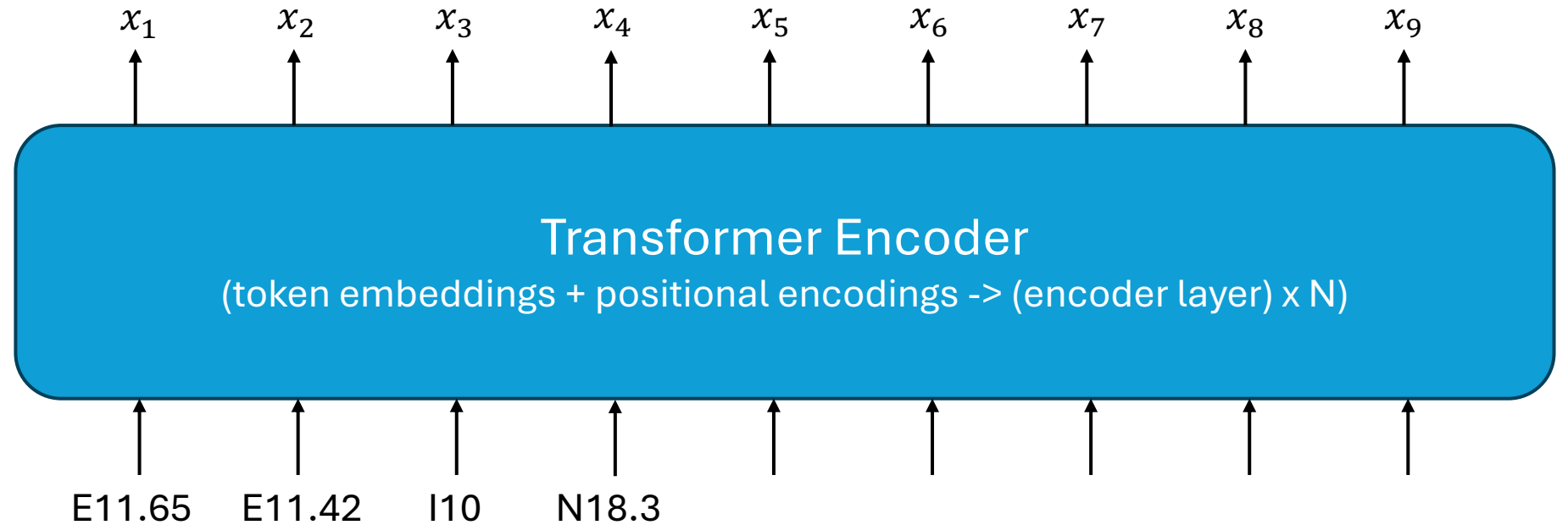
Output
representations



Transformers for Electronic Health Records

Unique consideration 2: some tokens have associated values (e.g., labs, vitals)

Output
representations



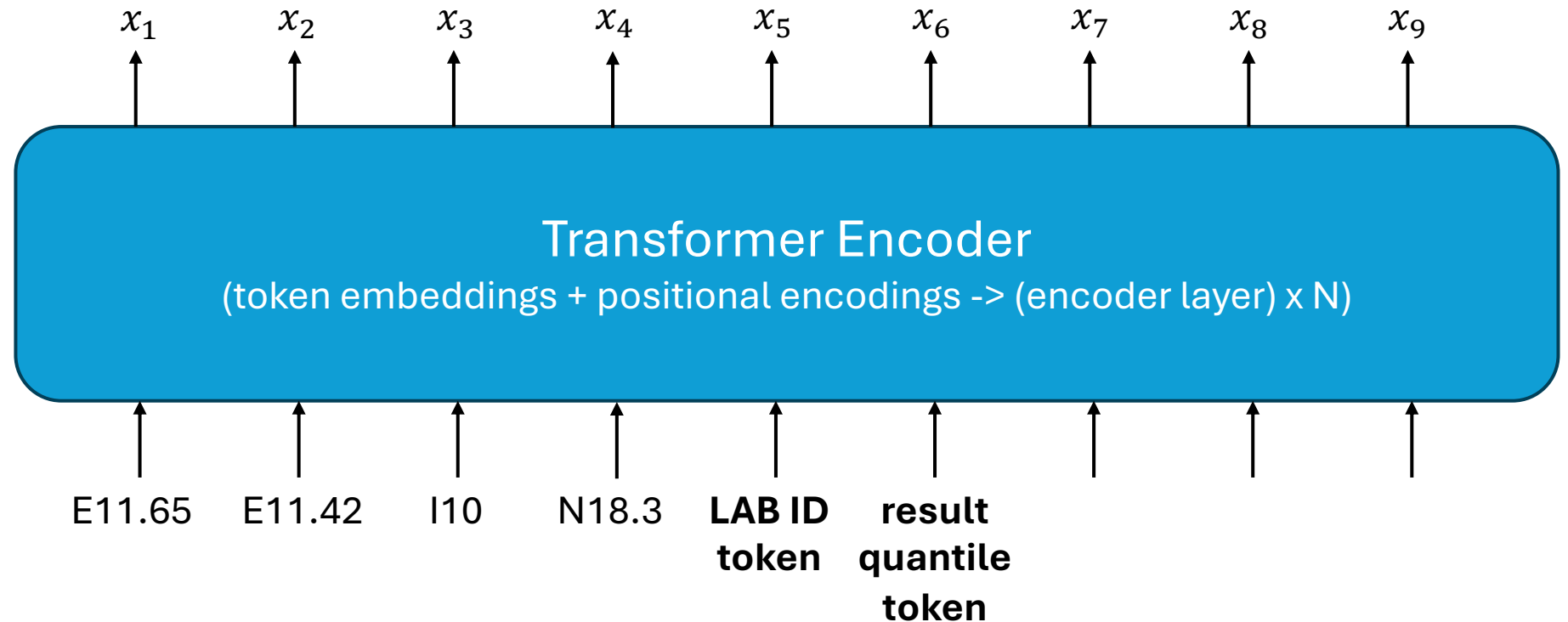
Input tokens

Transformers for Electronic Health Records

Unique consideration 2: some tokens have associated values (e.g., labs, vitals)

One approach to this is shown below

Output
representations

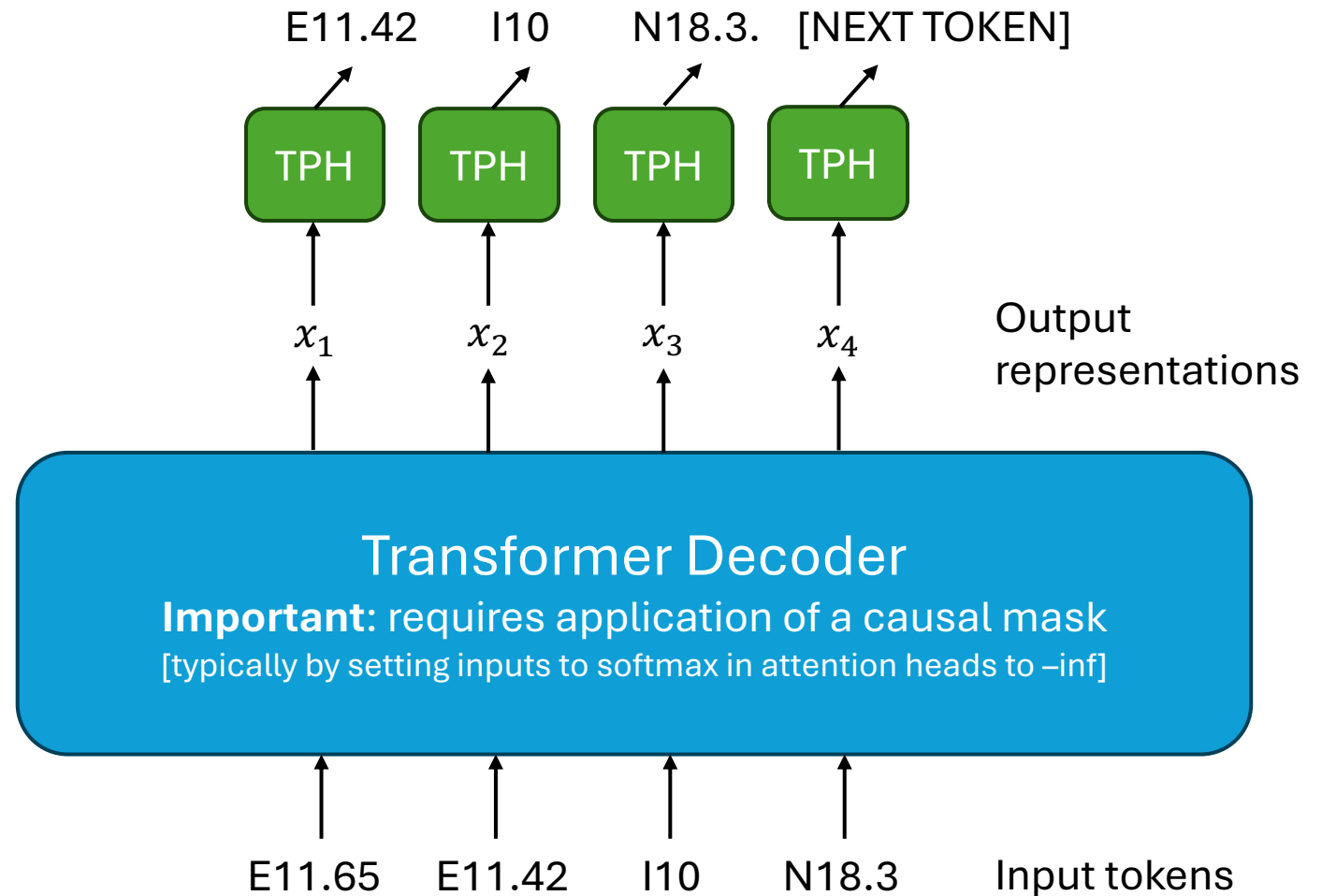


Predict diagnosis risk from structured EHRs

- Suppose we wish to predict which children are most likely to be diagnosed with asthma based on EHRs from early life
- How would we do this **without** a transformer-based foundation model?
- How would we do this **with** a transformer-based foundation model?
- What are the advantages of the latter approach?

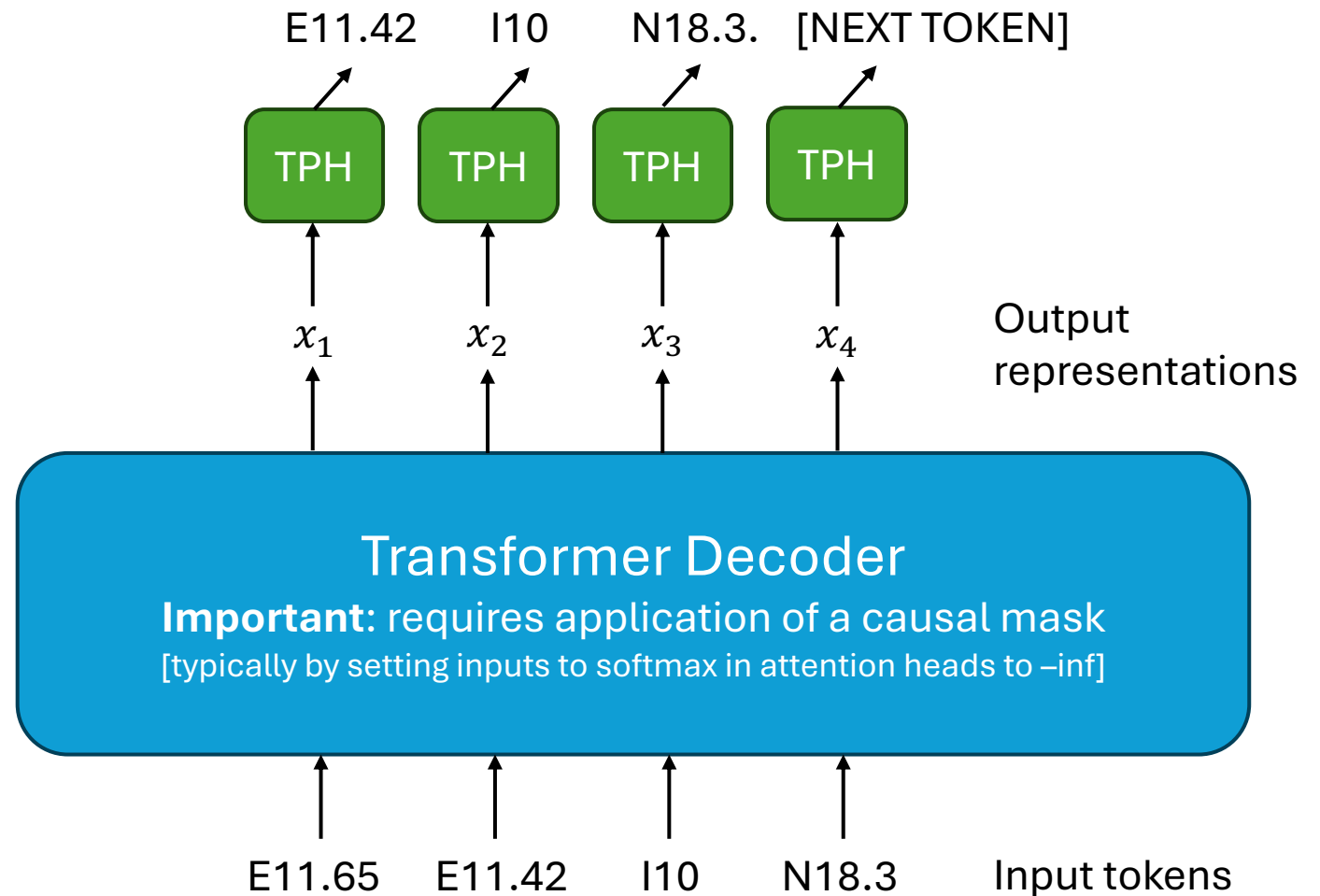
Zero-shot prediction via EHR generation

- Suppose we wish to predict which children are most likely to be diagnosed with asthma based on EHRs from early life
- We don't have any task-specific data; we only have a decoder-only foundation model trained to predict the next token/event
- How might we use this to perform zero-shot asthma prediction?



Zero-shot prediction via EHR generation

- Suppose we wish to predict which children are most likely to be diagnosed with asthma based on EHRs from early life
- How might we use this to perform zero-shot asthma prediction?
 1. Generate many sequences conditioned on the observed history
 2. Count how many sequences contain asthma
 3. Predicted prob is the proportion that contain asthma



What are the benefits of a *foundation model*?

- General-purpose, effective representations that can be used for a wide range of tasks
- Improved performance on downstream tasks
 - Many-shot
 - Few-shot
- Allows zero-shot prediction (similar to text models)

Do foundation models generalize?

Do we need these models to generalize?

- View 1: FM should work everywhere
- View 2: FM is a recipe that can be used anywhere

npj | digital medicine

Published in partnership with Seoul National University Bundang Hospital

Editorial



<https://doi.org/10.1038/s41746-026-02615-4>

The portability paradox of foundation models for clinical decision support

Check for updates

Yakdan et al. demonstrate that foundation models (FMs) trained to predict cervical spondylotic myelopathy from electronic health record data outperform traditional models on internal datasets but lose their advantage during external validation. This suggests that the feature-dense patterns learned by FMs may reduce their portability across settings, particularly for rare outcomes. As FMs approach clinical deployment, local validation, subgroup analysis, and attention to implementation burden are essential to inform health system planning and stewardship.

adapted to perform many specific tasks using smaller, curated datasets.

FMs thereby function as reusable infrastructure, offering several pragmatic advantages. Though upfront investment is significant, FM adaptation requires less labeled training data than building a new model from scratch, potentially reducing development time and costs in the long-term^{11–14}. Reduced labeling requirements also make it possible to build models when outcome data is limited, such as for rare disease detection or at smaller institutions. Operationally, managing a shared, multi-use platform may be simpler than maintaining numerous independent models.

Evidence from imaging, language, and other domains suggests that FMs can outperform task-specific models and generalize better across settings and time^{15–19}. Early studies of EHR-based

evaluated on Merative data. These values look small, but reflect the low prevalence of CSM: an AUPRC of 0.358 is a 15-fold improvement over random prediction. In the same experiment, a smaller, clinically informed model scored a mean AUPRC of 0.137 internally and 0.110 externally.

A reasonable explanation is that FMs are prone to overfitting, meaning they learn to make predictions based on patterns in a training dataset that are not present or helpful in another dataset²⁵. For instance, after training with one institution's data, an FM might learn to predict CSM diagnosis when patients receive a spine MRI then surgical consultation within three months. If applied to another hospital with longer wait times or a preference for surgical consultation before imaging, this pattern becomes useless and the model's performance will suffer. Simpler models trained with less data are worse at picking up such

Why multimodal fusion?

What will we find in the structured data?

- Encounter details (ED admission)
 - Labs (BMP, CBC)
 - Vital signs
 - Procedure codes (CT)
 - Diagnosis codes (PE)
-
- ...what *won't* we find?

Why do we need many modalities?



Imaging

PCOR_RSH_DIAGNOSIS_VW EXAMPLE																
DIAGNOSISID	PATID	ENCOUNTERID	ENC_TYPE	ADMIT_DATE	PROVIDERID	DX	DX_TYPE	DX_SOURCE	DX_ORIGIN	PDX	DX_POA	RAW_DX	RAW_DX_TYPE	RAW_DX_SOURCE	RAW_PDX	
36510180	123456	57896636	AV	2014/02/21 00:00:00	1745326	530.9	9	FI	BI	OT	NI	530.9	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY	
36510407	223456	57897312	AV	2014/05/07 00:00:00	73411	530.81	9	FI	BI	OT	NI	530.81	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY	
36510405	323456	57894371	ED	2014/06/13 00:00:00	123901	553.3	9	FI	BI	OT	NI	553.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY	
26510406	423456	57843289	AV	2015/01/21 00:00:00	1719263	530.3	9	FI	BI	OT	NI	530.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY	
36510721	523456	57898224	AV	2015/03/04 00:00:00	1719263	787.2	9	FI	BI	OT	NI	787.2	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY	
36510720	612345	57899120	ED	2015/07/30 00:00:00	156342	454.8	9	FI	BI	OT	NI	454.8	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY	
36511049	654321	57892268	EI	2016/12/03 00:00:00	165493	V12.72	9	FI	BI	OT	NI	V12.72	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY	
36508682	543216	57893648	IP	2016/08/05 00:00:00	178238	211.3	9	FI	BI	OT	NI	211.3	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY	

Structured EHR data

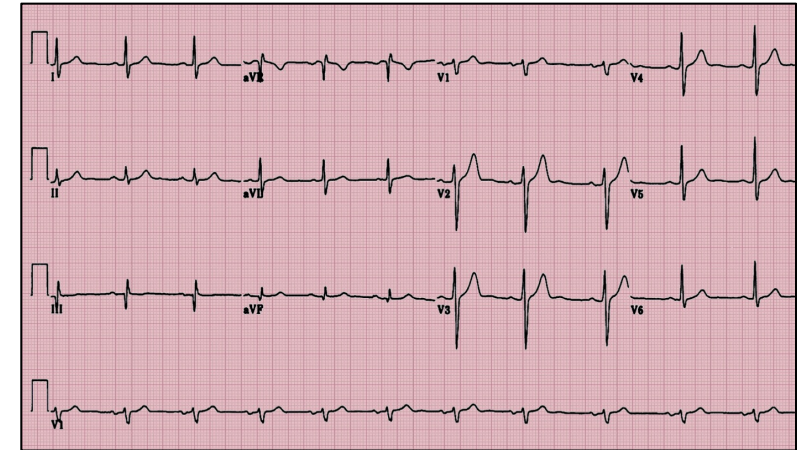
Clinical notes

Chief Complaint:

Shortness of breath.

History of the Present Illness:

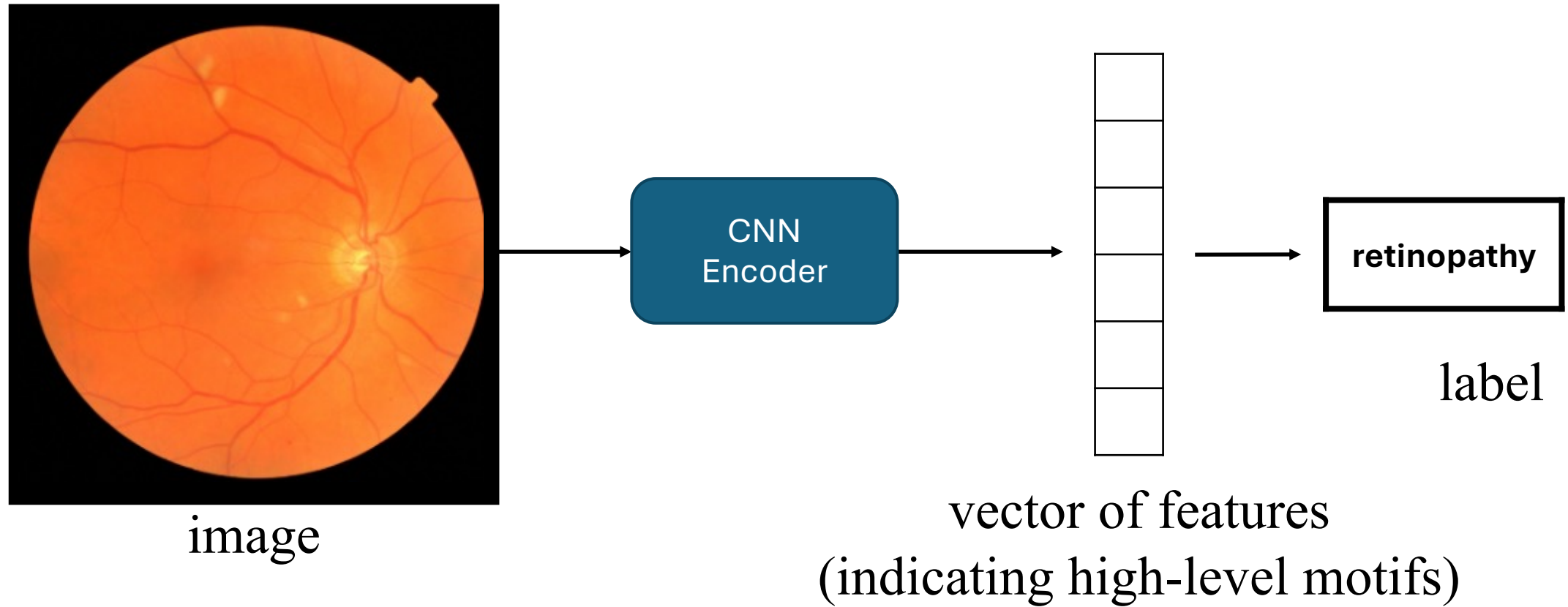
Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and



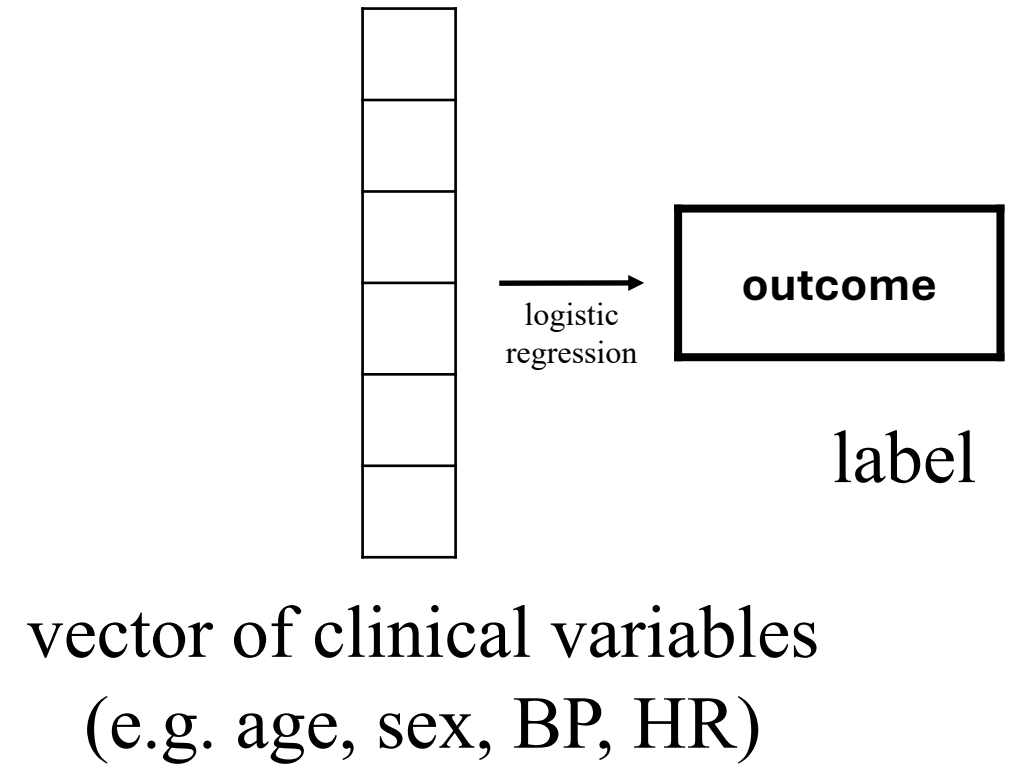
EKG and other physiologic time-series

How do we do multimodal
fusion?

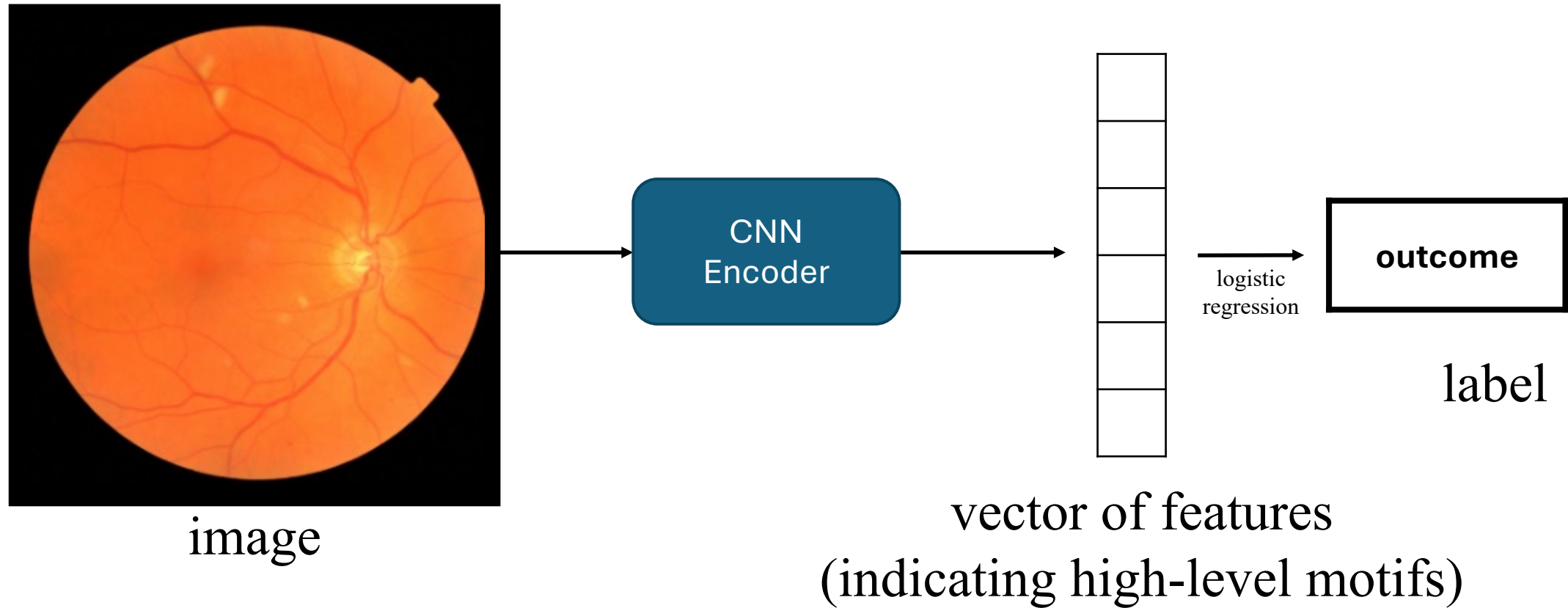
Image Encoder



A typical prediction model



A prediction model for images



A prediction model for documents

Chief Complaint:
Shortness of breath.

History of the Present Illness:
Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and Benzonatate 200 mg TID, which he has been taking for two days with only slight improvement. He then drove from Abilene back to Greensboro, where he resides, and continued to experience shortness of breath, right sided chest pain, and hemoptysis. He presented to an urgent care office in town today, and was subsequently transferred to the Moses Cone ER due to the provider's suspicion of PE.

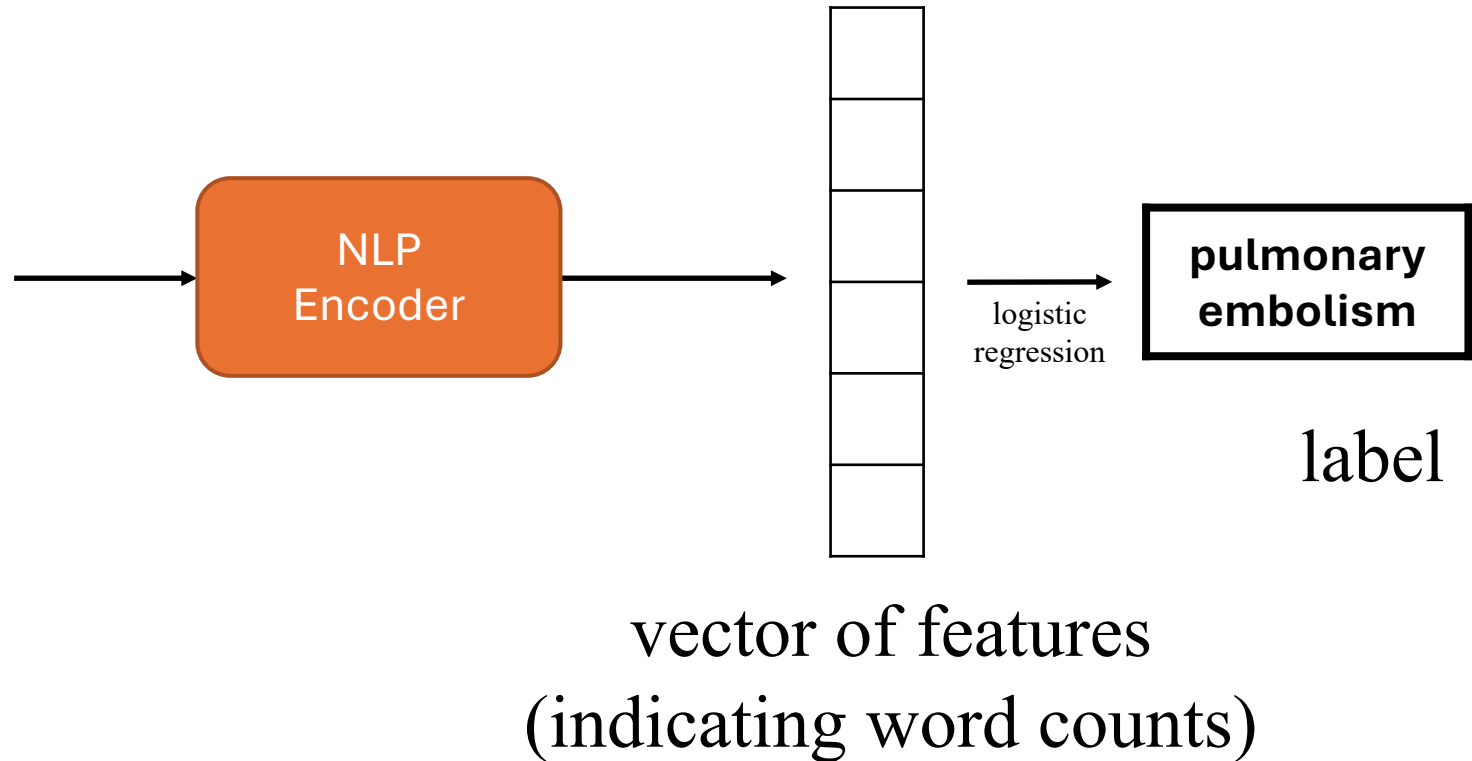
The right-sided pain is located midway down his ribcage, below the axilla. This pain is sharp, about 7/10 in severity, and worsens with movement and cough. Pressing on the chest does not recreate the pain. He feels that the pain has improved somewhat over the past two days. The hemoptysis has been unchanged since it began; there is not frank blood, but his sputum has been consistently blood-tinged. The blood seems redder at night. The dyspnea has been severe, and it is difficult for him to walk more than across a room. He states that he feels as though there is a "rattling" in his chest. At baseline, he experiences no dyspnea on exertion and has no history of COPD or other respiratory problem. He is a smoker, smoking a little less than a pack a day for thirty-five years. Past history is notable for the fact that he experienced transient left lower leg swelling – from below the knee down – and pain several weeks ago during a cross-country haul. He also notes a four day history of decreased appetite, poor sleep, and subjective fever and chills, with a measured fever of 103 in the hospital in Abilene. He had a bout of pneumonia about two months ago, but has been healthy for the most part and denies any chronic medical conditions. Currently he is fairly comfortable, with morphine helping with the pain. He has no history of a clotting disorder, no cardiac history, and denies any chest trauma or aspiration. He has had no sick contacts.

Past Medical History:

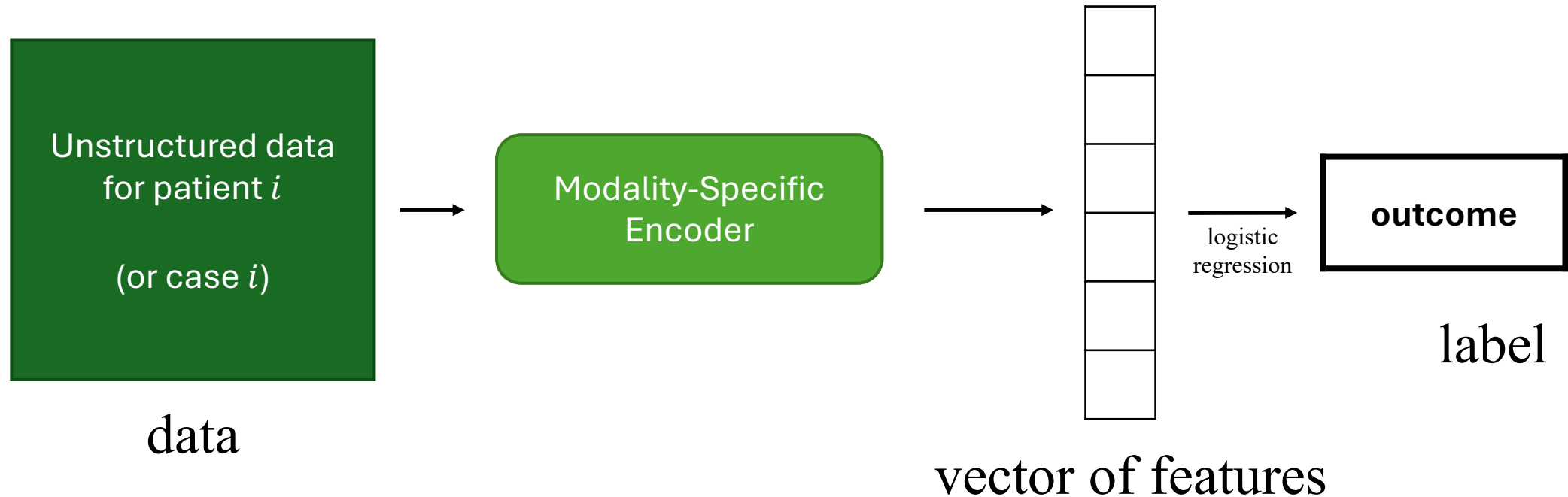
1. Hernia repair
2. Bilateral thumb surgeries, secondary to two separate injuries sustained while working with machinery

Medications:
No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Abilene: Levaquin 500 mg daily and Benzonatate 200 mg TID.

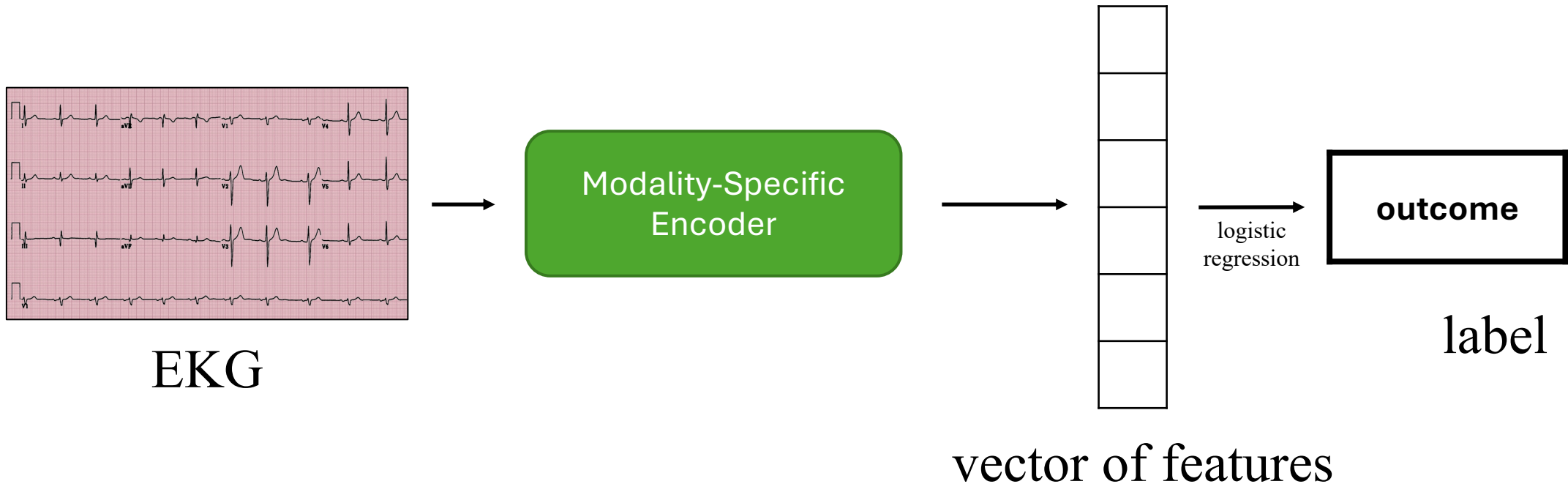
document



A prediction model for unstructured data

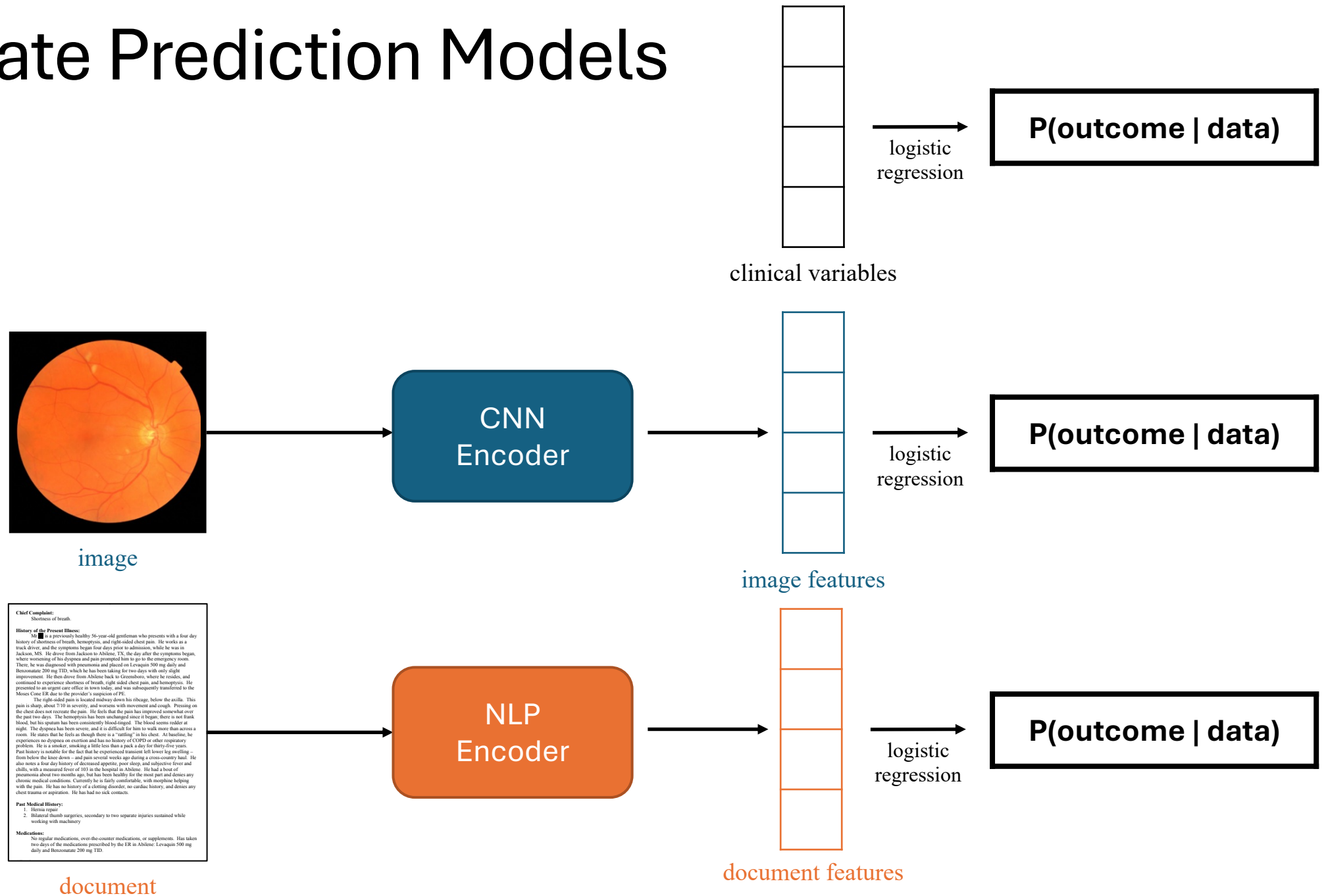


A prediction model for unstructured data



1. Late fusion

Separate Prediction Models



Late fusion



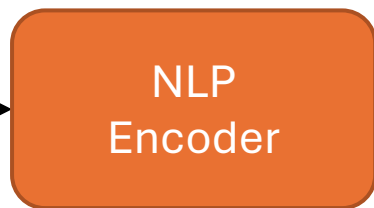
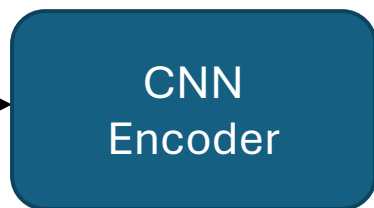
image

at-old gentleman who presents with a four day
and right-sided chest pain. He works as a
days prior to admission, while he was in
bilious. 12, the day after the symptoms began,
prompted him to go to the emergency room.
and placed on Levaquin 500 mg daily and
en taking for two days with only slight
back to Greensboro, where he resides, and
right-sided chest pain, and hemoptysis. He
today, and was subsequently transferred to the
room of PE.
tway down his ribcage, below the axilla. This
resolves with movement and cough. Pressing on
ds that the pain has improved somewhat over
to unchanged since it began, there is not flank
blood-tinged. The blood seems redder at
is difficult for him to walk more than across a
there is a "rattling" in his chest. At baseline, he
is no history of COPD or other respiratory
less than a pack a day for thirty-five years.
experienced transient left lower leg swelling
-ut weeks ago during a cross-country haul. He
poor sleep, and subjective fever and
hospital in Asheville. He had a bout of
been healthy for the most part and denies any
fairly comfortable, with morphine helping
g disorder, no cardiac history, and denies any
sick contacts.

y to two separate injuries sustained while

ounter medications, or supplements. Has taken
bed by the ER in Asheville: Levaquin 500 mg

document



clinical variables

image features

document features



logistic
regression



logistic
regression



logistic
regression

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

aggregate predictions

Late fusion



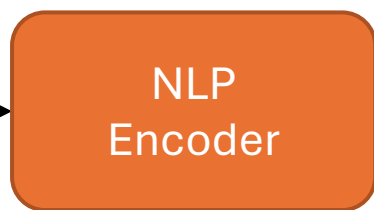
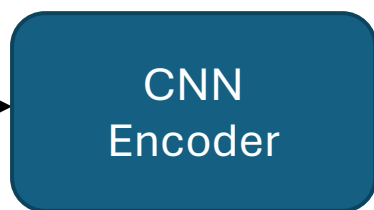
image

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peptic, poor sleep, and subjective fever and
hospital in Chile. He had a bout of
been healthy for the most part and denies any
fairly comfortable, with morphine helping
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sick contacts.

y to two separate injuries sustained while

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bed by the ER in Chile: Levaquin 500 mg

document



clinical variables

image features

document features



logistic
regression



logistic
regression



logistic
regression

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

aggregate predictions

- Average

(good default strategy;
assumes all modalities
are equally informative)

Late fusion



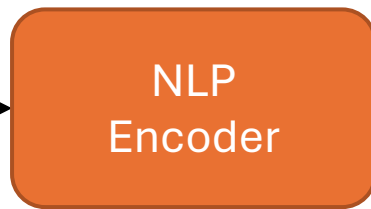
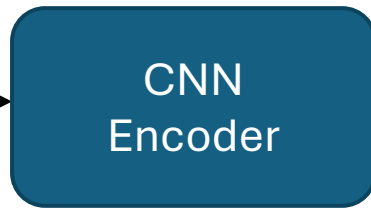
image

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clinical variables

image features

document features



logistic
regression



logistic
regression



logistic
regression

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

aggregate predictions

- Average
- Max

(used in multiple-
instance learning;
makes sense when the
relevant signal might be
present only in a single
modality)

Late fusion



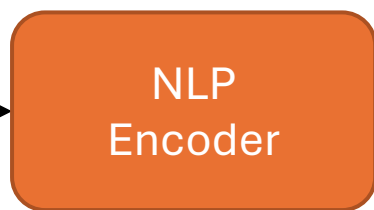
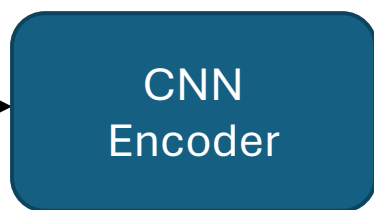
image

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image features

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logistic
regression



logistic
regression



logistic
regression

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

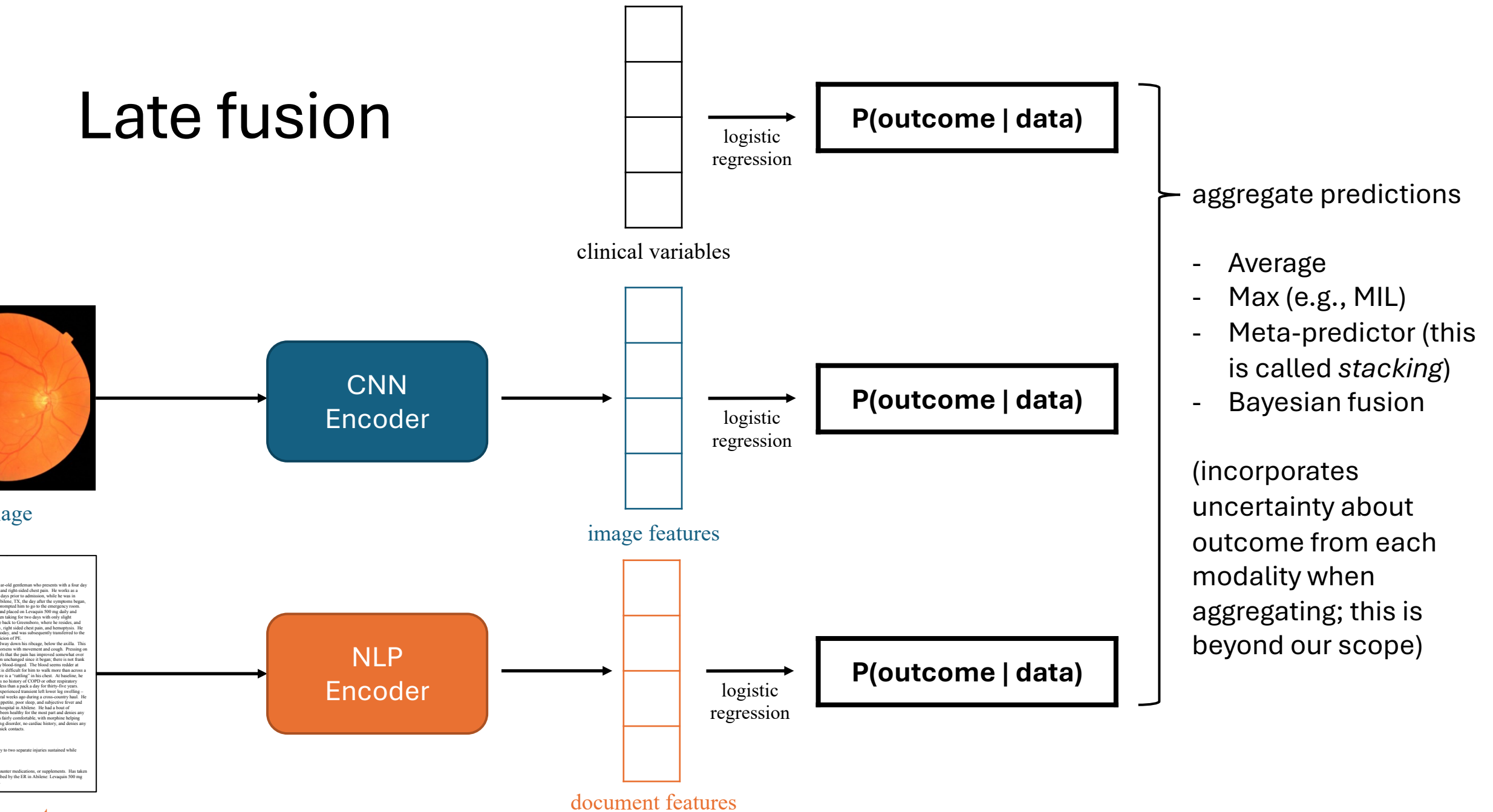
$P(\text{outcome} \mid \text{data})$

aggregate predictions

- Average
- Max (e.g., MIL)
- Meta-predictor (this is called *stacking*)

(flexible approach but
does require additional
training, at which point
joint fusion might be
better)

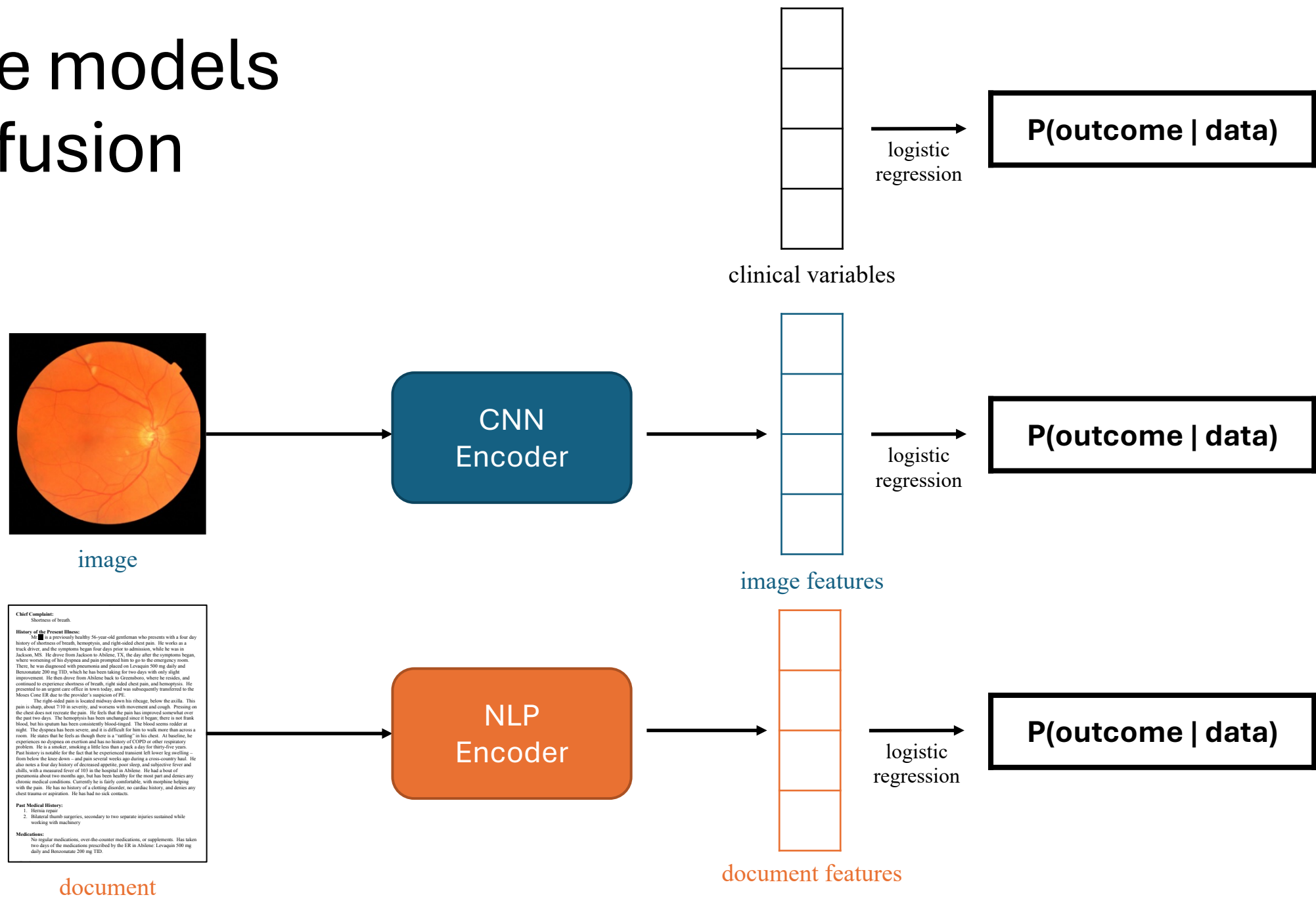
Late fusion



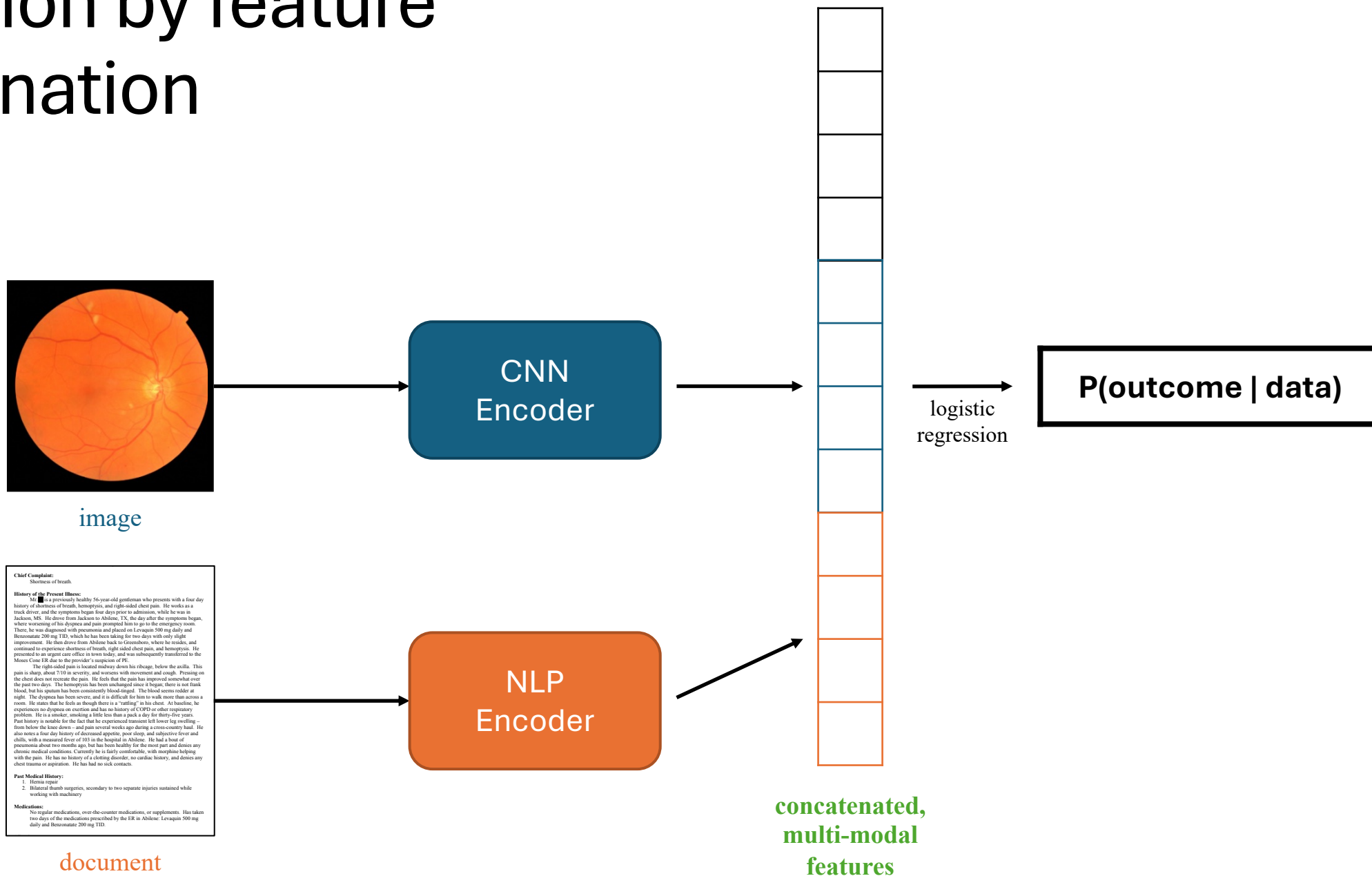
2. Joint fusion

Separate models

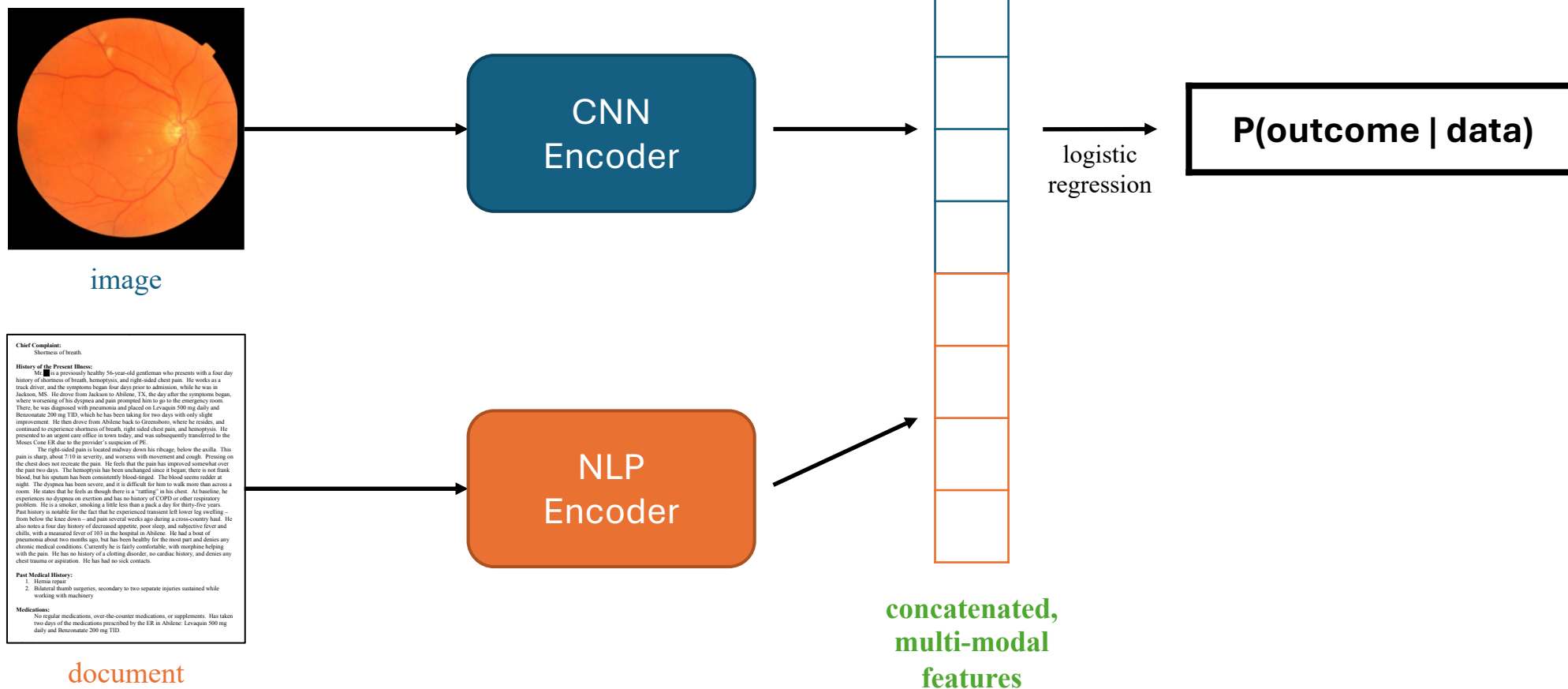
-> Joint fusion



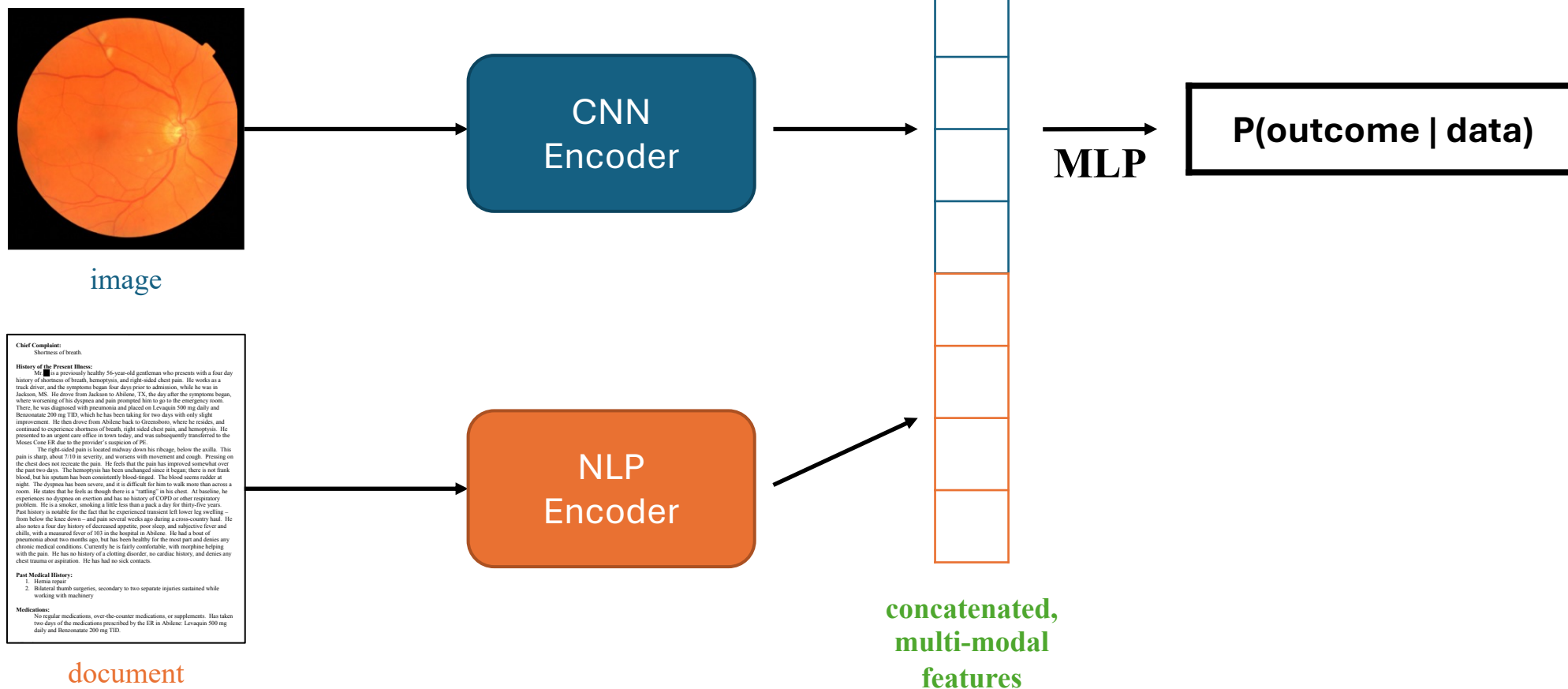
Joint fusion by feature concatenation



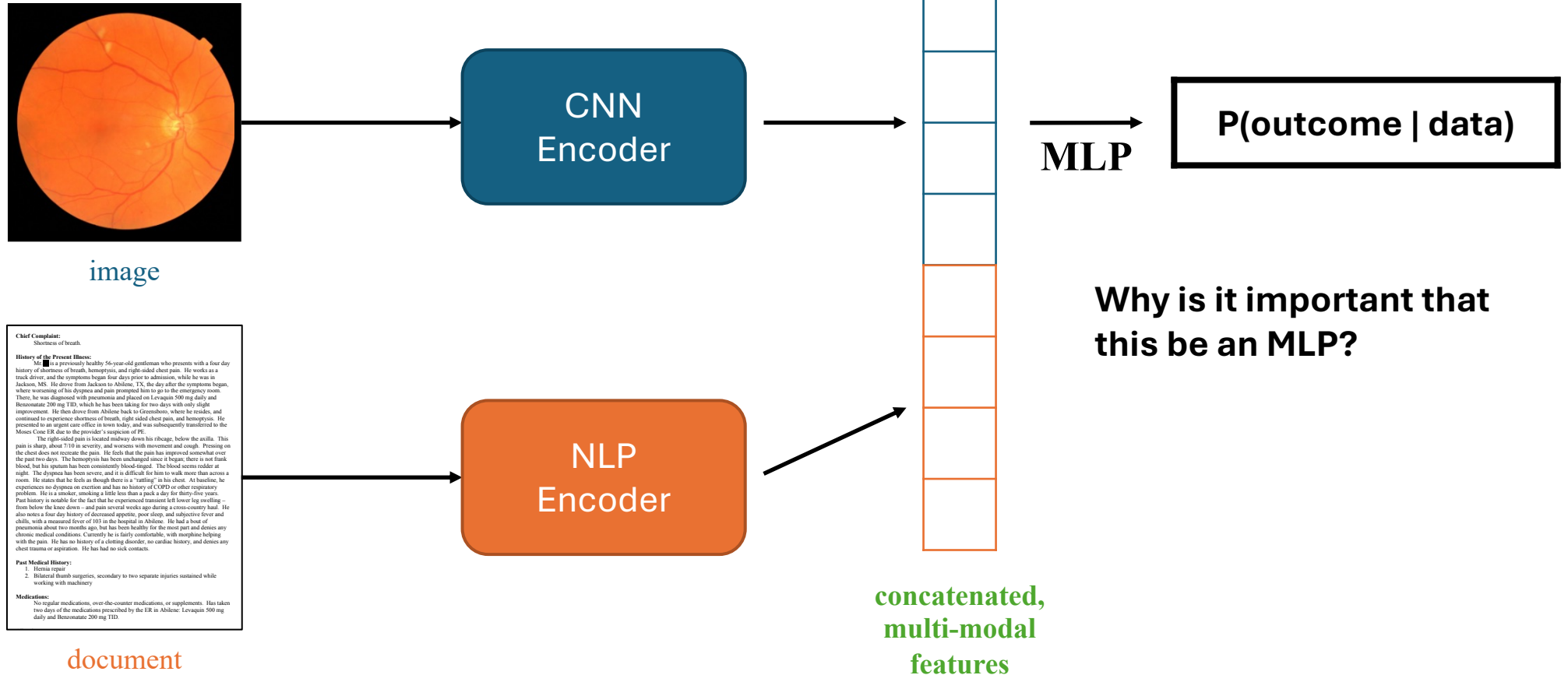
Joint fusion by feature concatenation



Joint fusion by feature concatenation

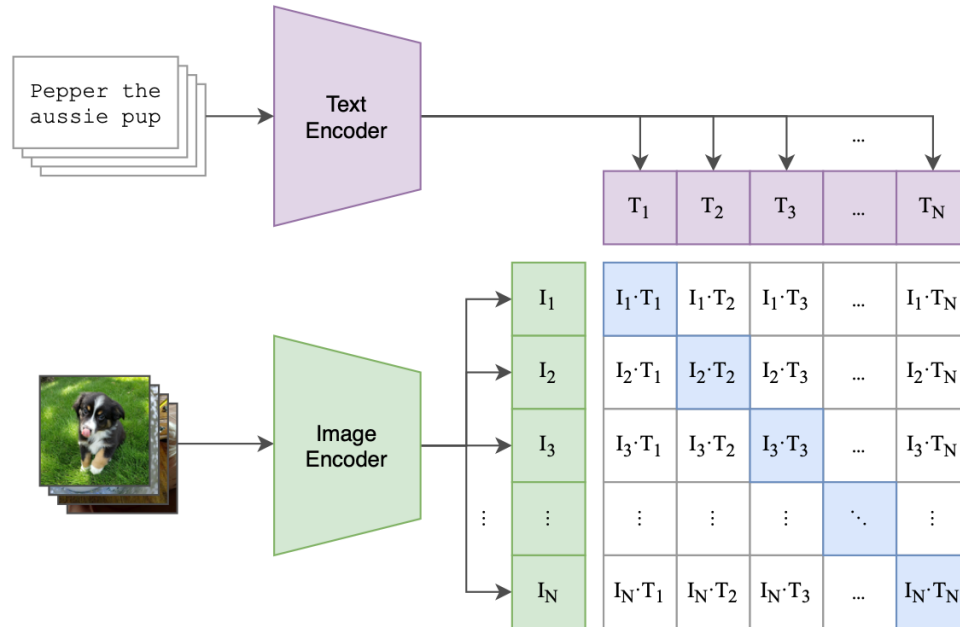


Joint fusion by feature concatenation



Joint fusion via contrastive learning

(1) Contrastive pre-training



(2) Create dataset classifier from label text

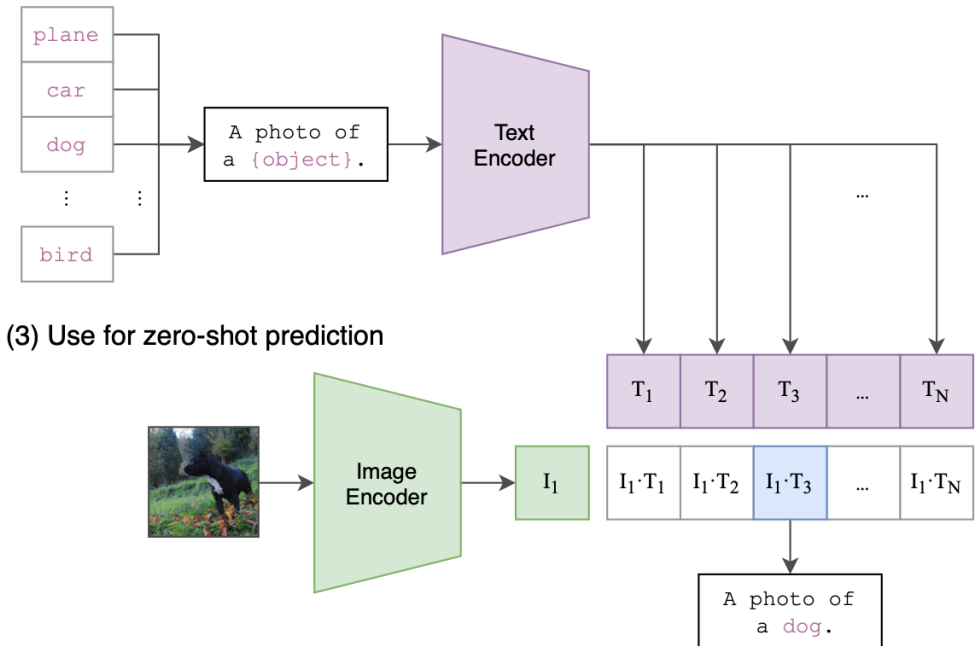
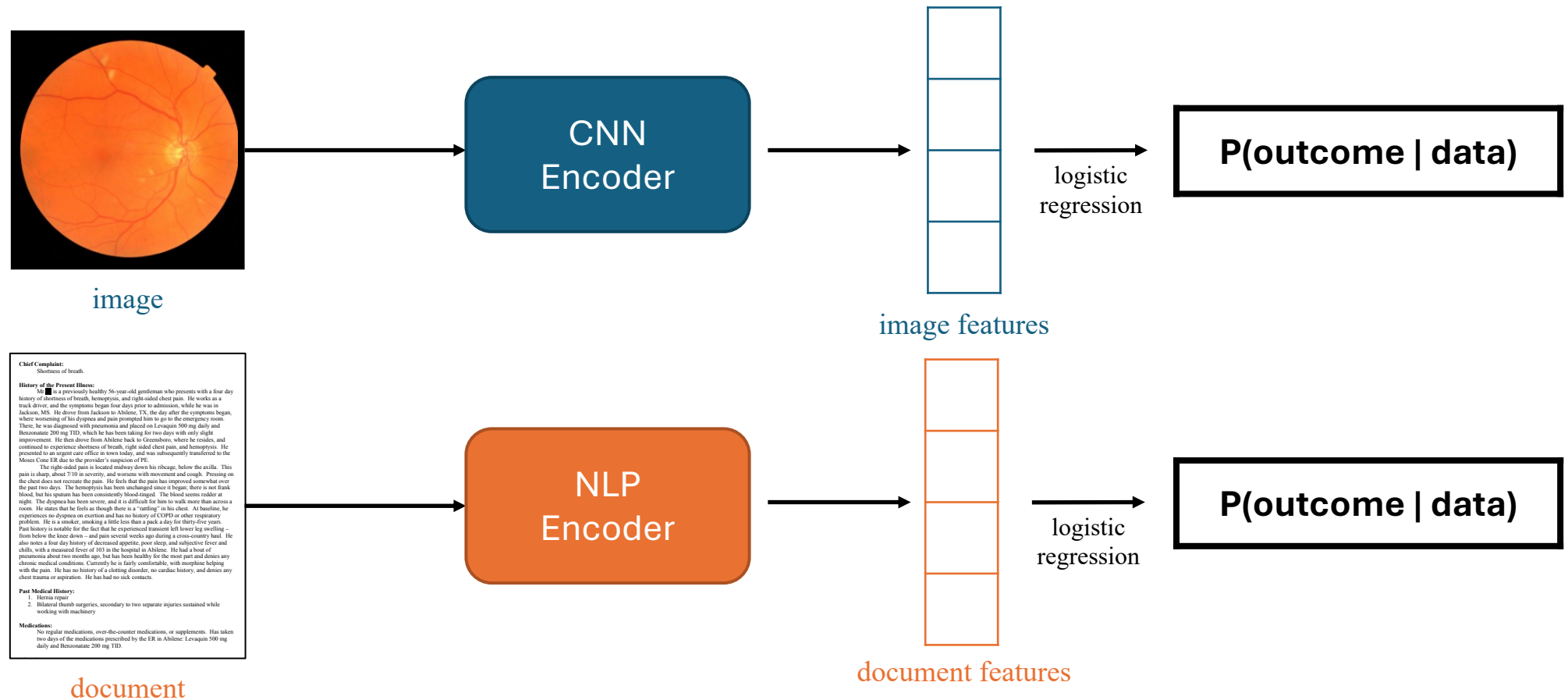
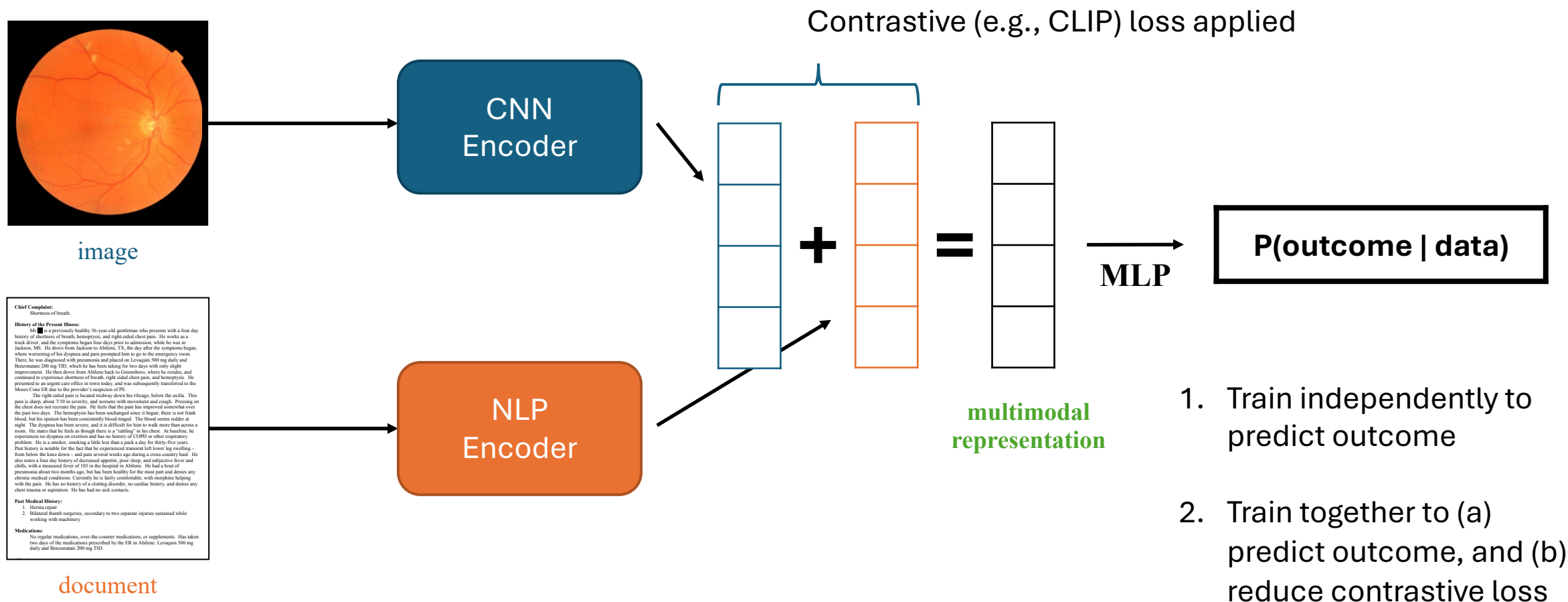


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Separate Prediction Models

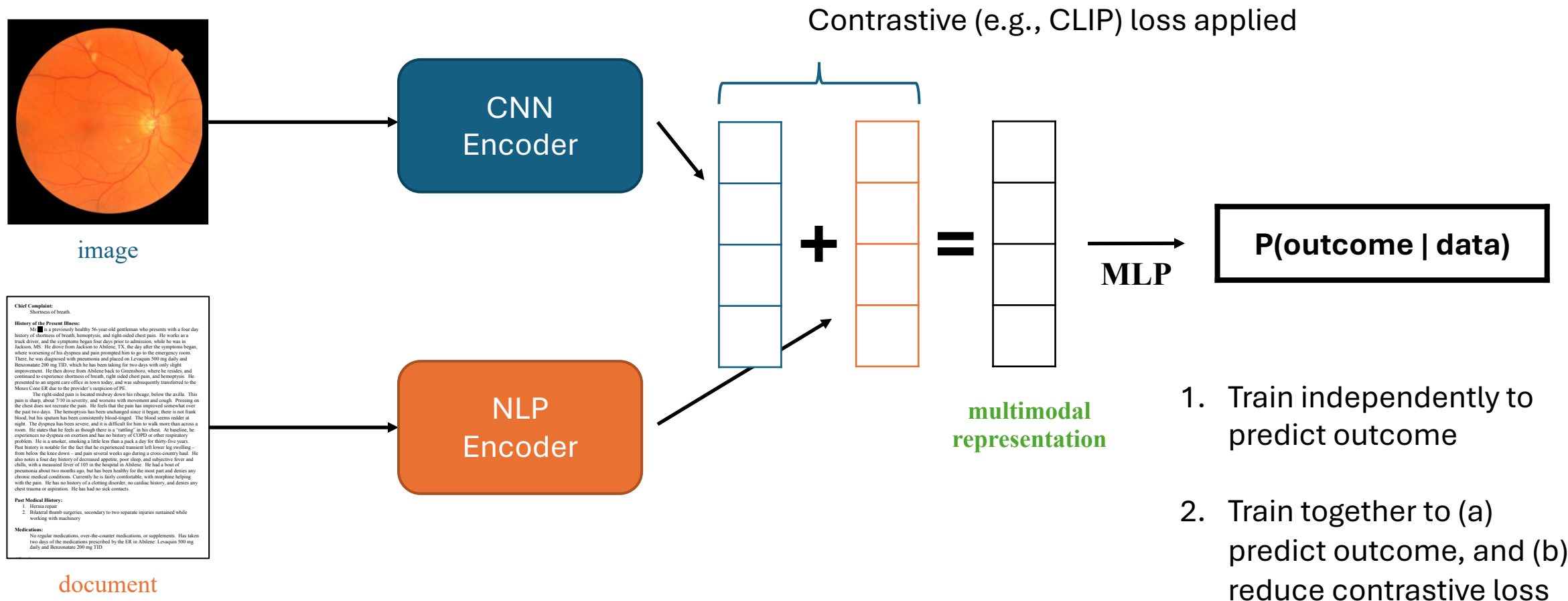


Joint fusion in shared representation space



Joint fusion in shared representation space

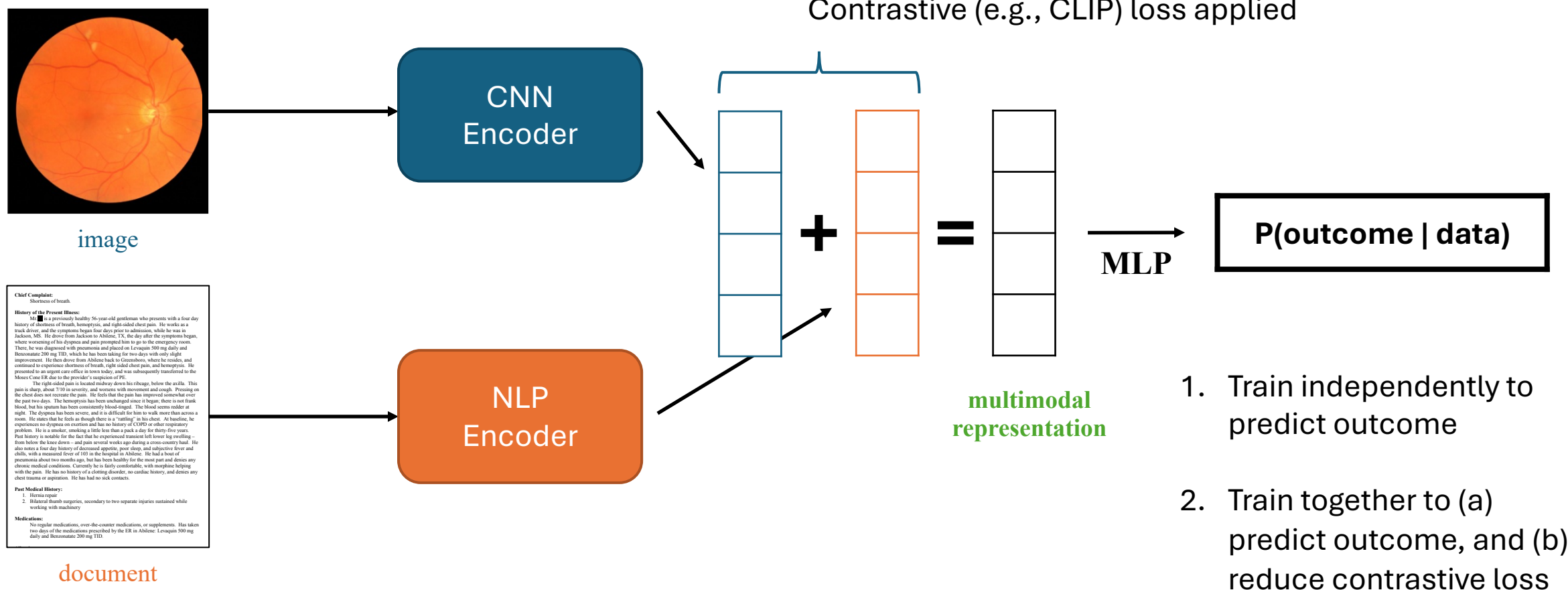
Q: What happens if we train to minimize the contrastive loss only?



Joint fusion in shared representation space

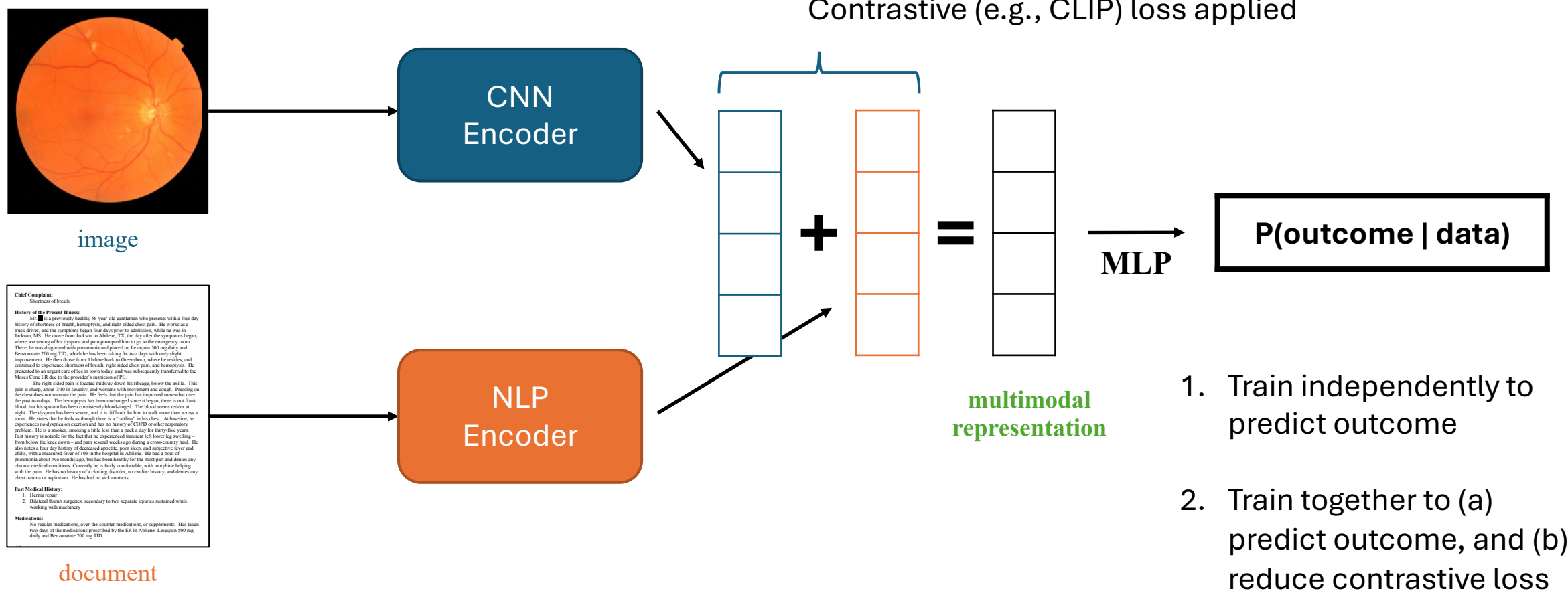
Q: What happens if we train to minimize the contrastive loss only?

A: we will lose *distinct* information between modalities that is predictive of the outcome



Joint fusion in shared representation space

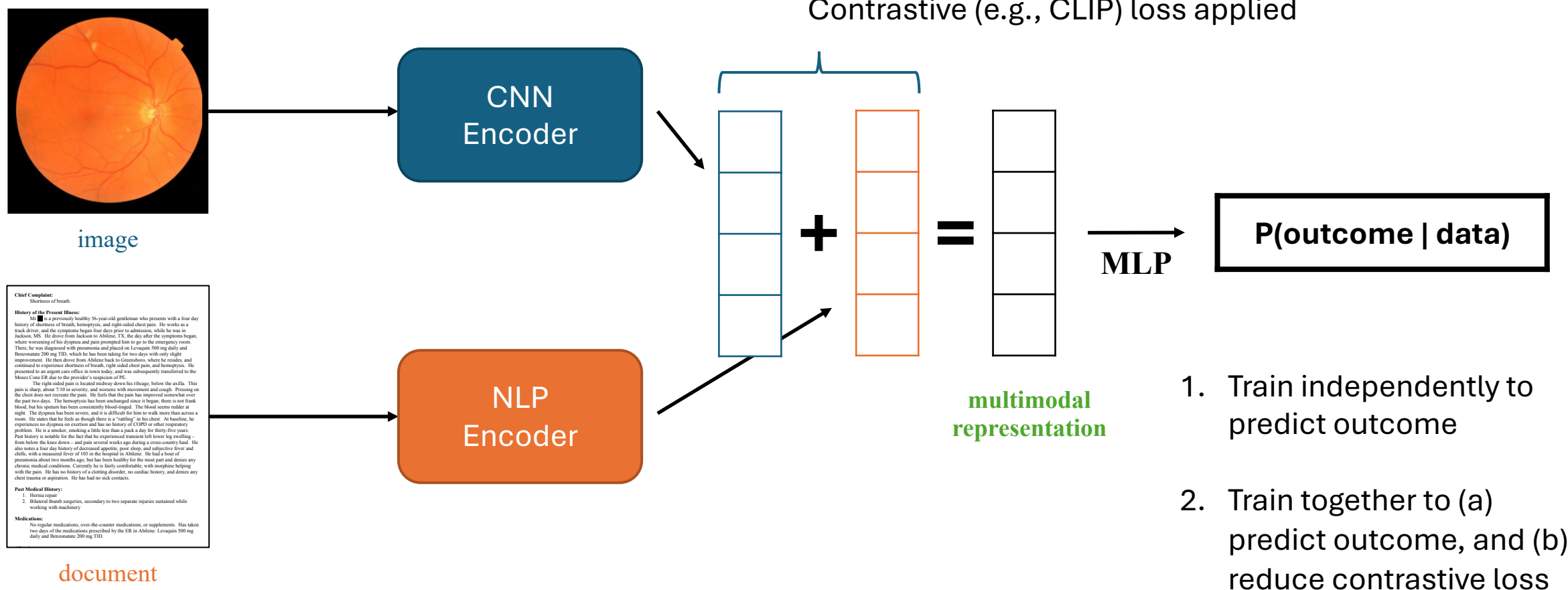
Q: Does this allow interactions between modalities?



Joint fusion in shared representation space

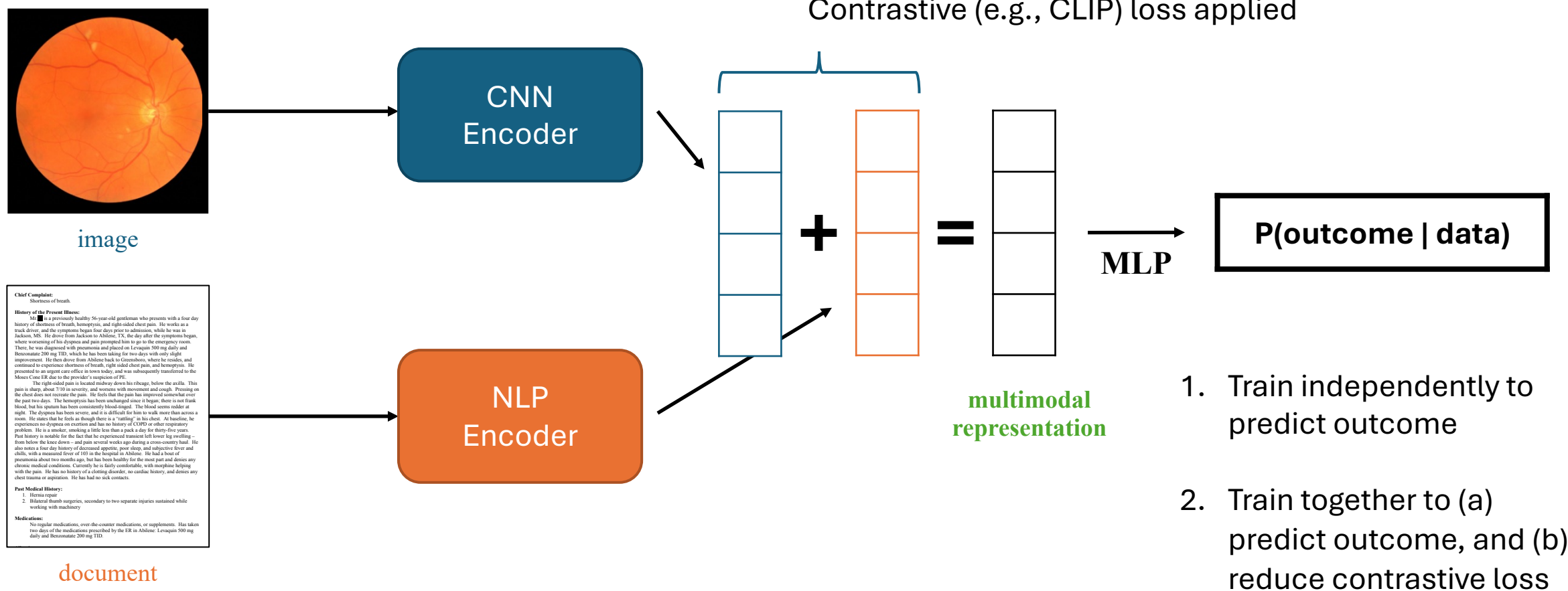
Q: Does this allow interactions between modalities?

A: Yes, because distinct information between modalities can go in separate channels



Joint fusion in shared representation space

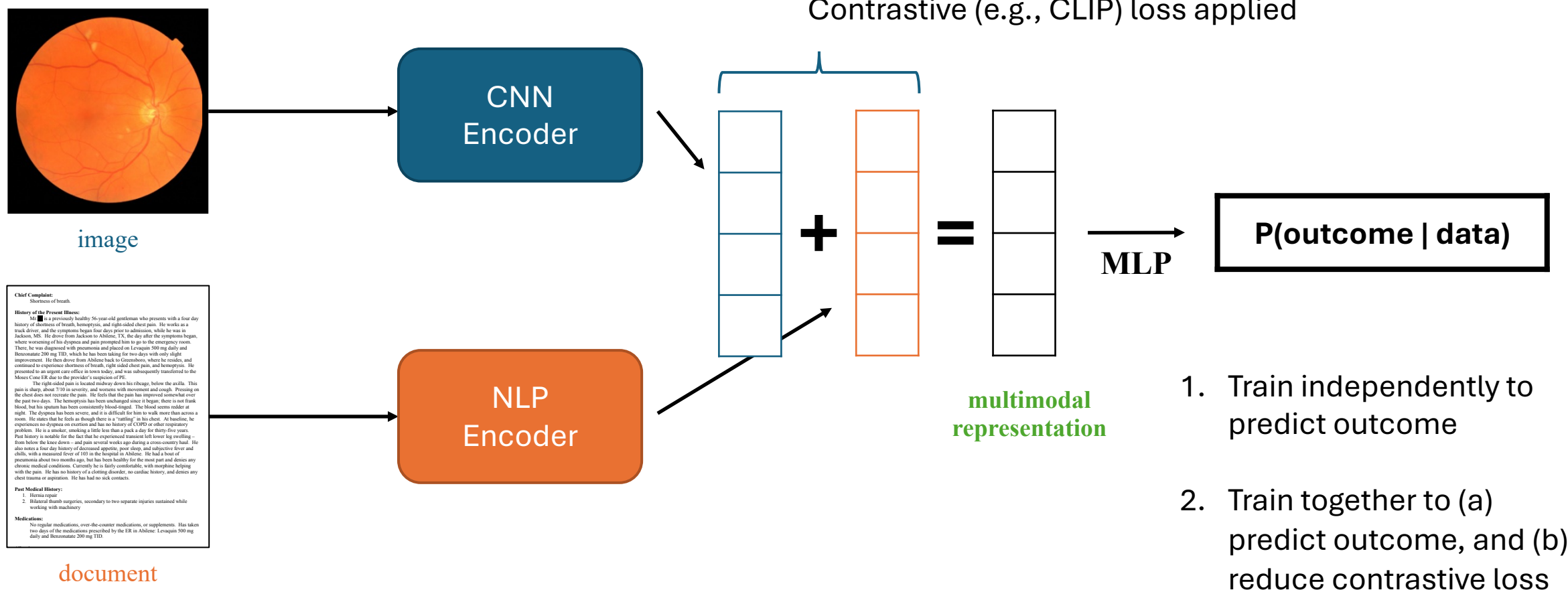
Q: Does adding result in loss of information?



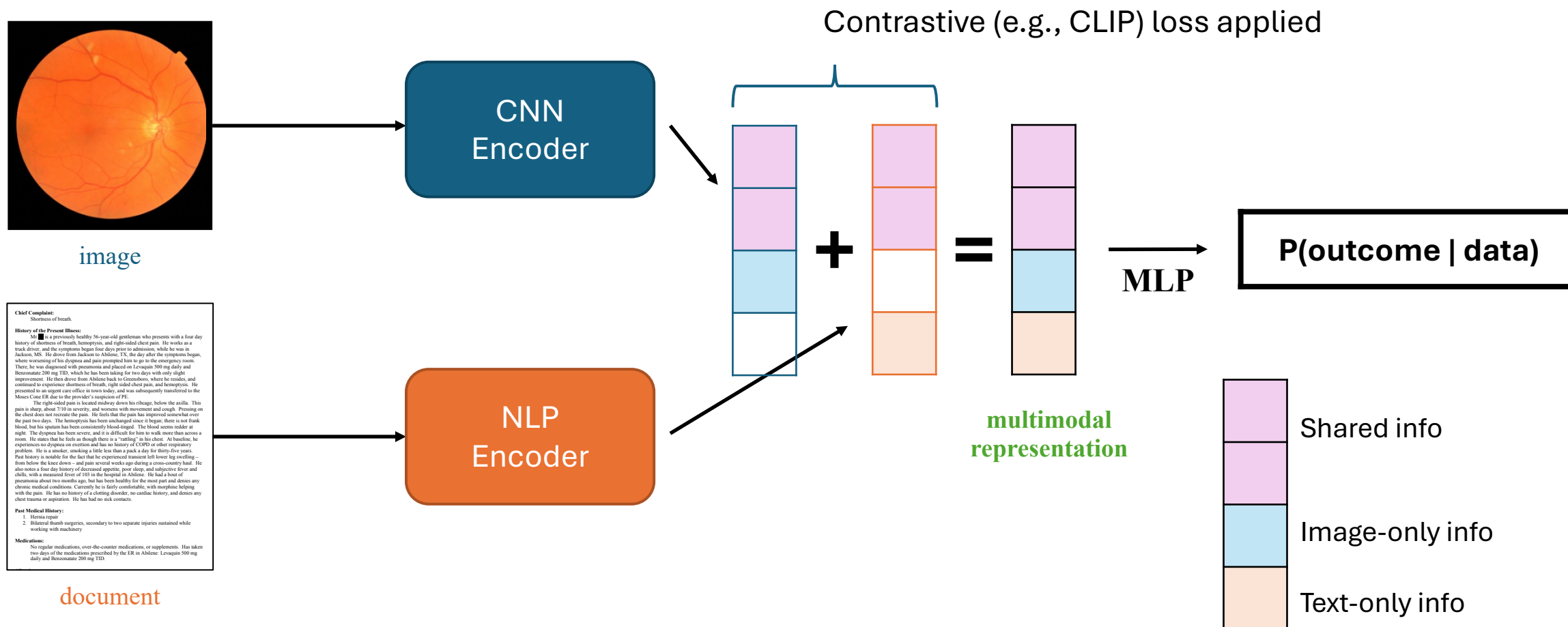
Joint fusion in shared representation space

Q: Does adding result in loss of information?

A: Not necessarily, because distinct information can go in separate channels.
Prediction loss helps ensure info is retained.

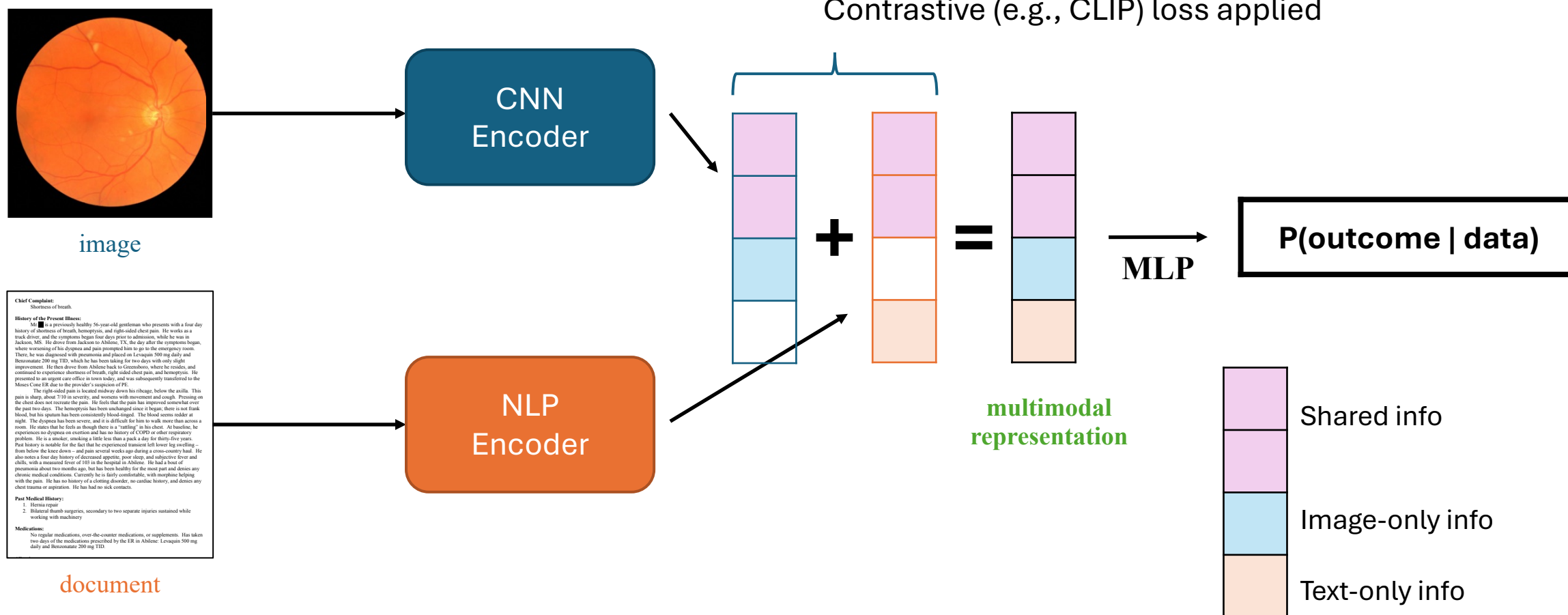


Joint fusion in shared representation space



Joint fusion in shared representation space

1. Contrastive loss encourages shared info between modalities to concentrate in a subset of channels
2. Prediction loss prevents distinct info from being lost



What happens with >2 modalities?

1. Define CLIP loss for each pair of modalities (usually fine)
2. Define a *primary* modality
 - Depends on the nature of the problem
 - Works best if one modality is particularly information rich; often text
3. Consider alternative loss structures
 - For example, [Symile-MIMIC](#) considers pairwise and higher-order information
 - Training objective maximizes a lower bound on the total correlation
 - Number of loss terms scales linearly rather than quadratically in the number of modalities

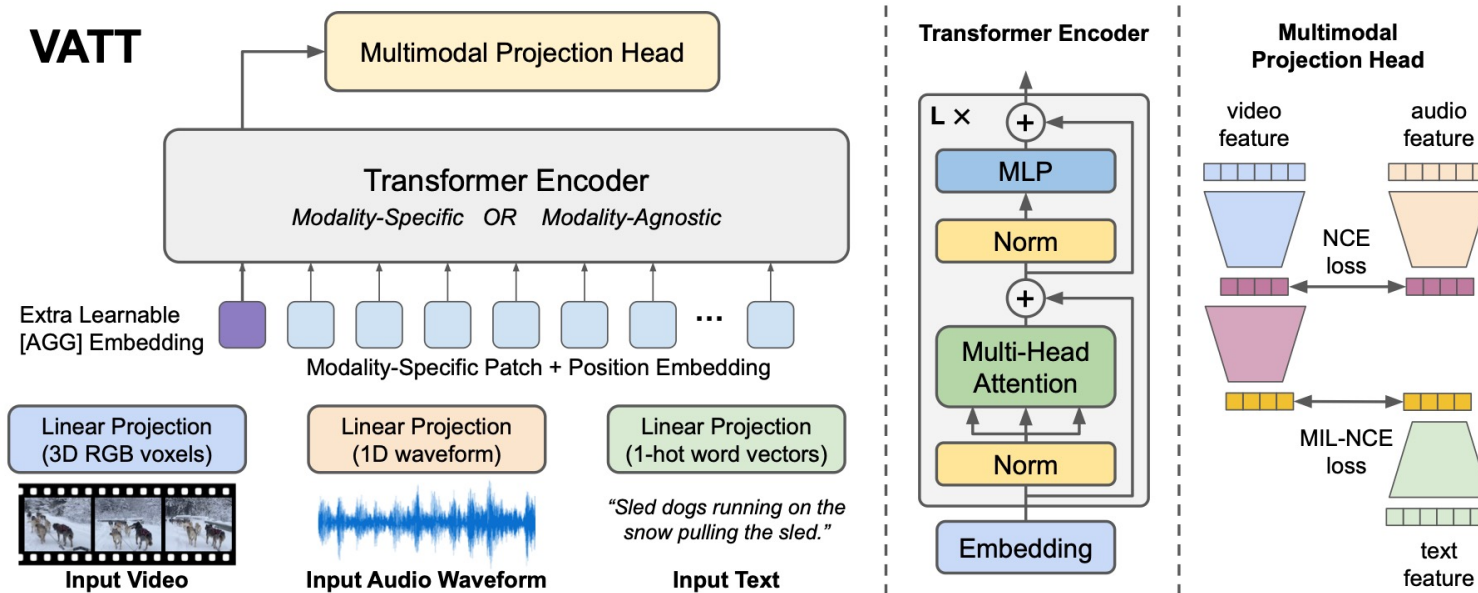
On interactions between modalities

- Late fusion does not allow interactions!
- Joint fusion does allow interactions, but flexibility is limited, as this happens only at the end of feature extraction
- Recall: this is similar to the concern that motivated the Transformer, namely that some interactions* were only possible in later layers of the model

*namely, long-distance interactions

3. Early fusion

Frontier models have moved to early fusion



- Distinct linear projections map raw pixels and waveforms to a shared input embedding space
- Contrastive loss is applied to encourage output representations to be similar across modalities

Akbari, H., Yuan, L., Qian, R., Chuang, W.-H., Chang, S.-F., Cui, Y., & Gong, B. (2021). VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text. *Advances in Neural Information Processing Systems (NeurIPS)* 34.

Comparing fusion methods

Late fusion

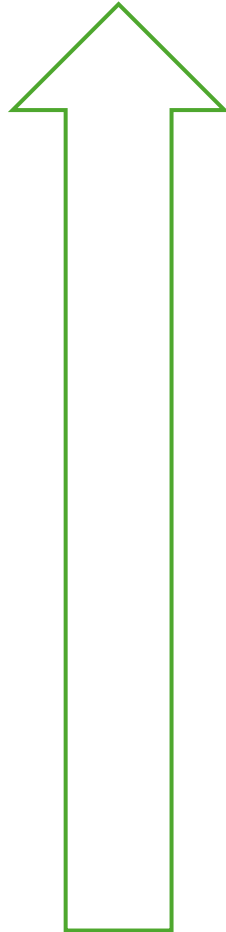
easiest

Joint fusion

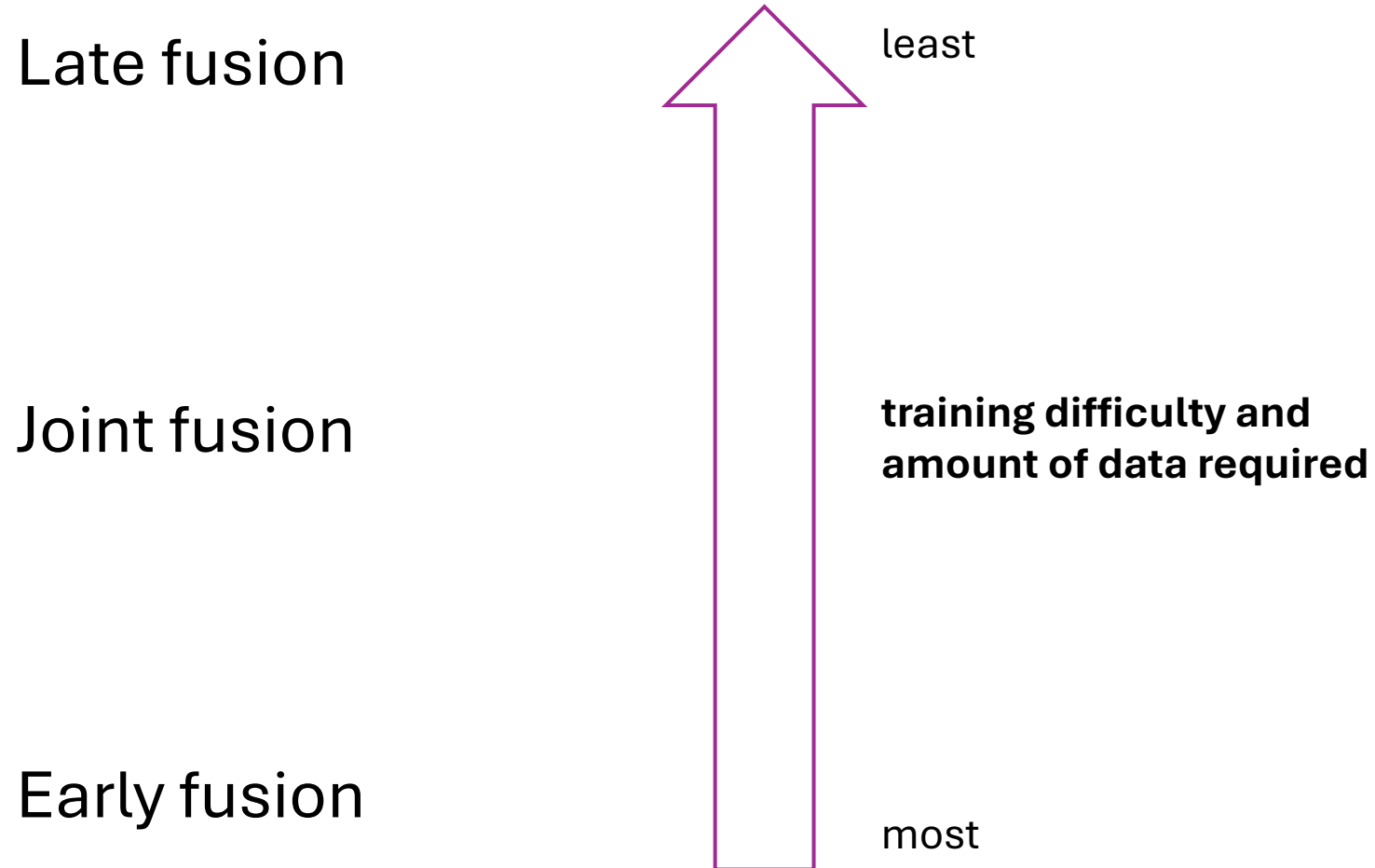
**ease of
implementation**

Early fusion

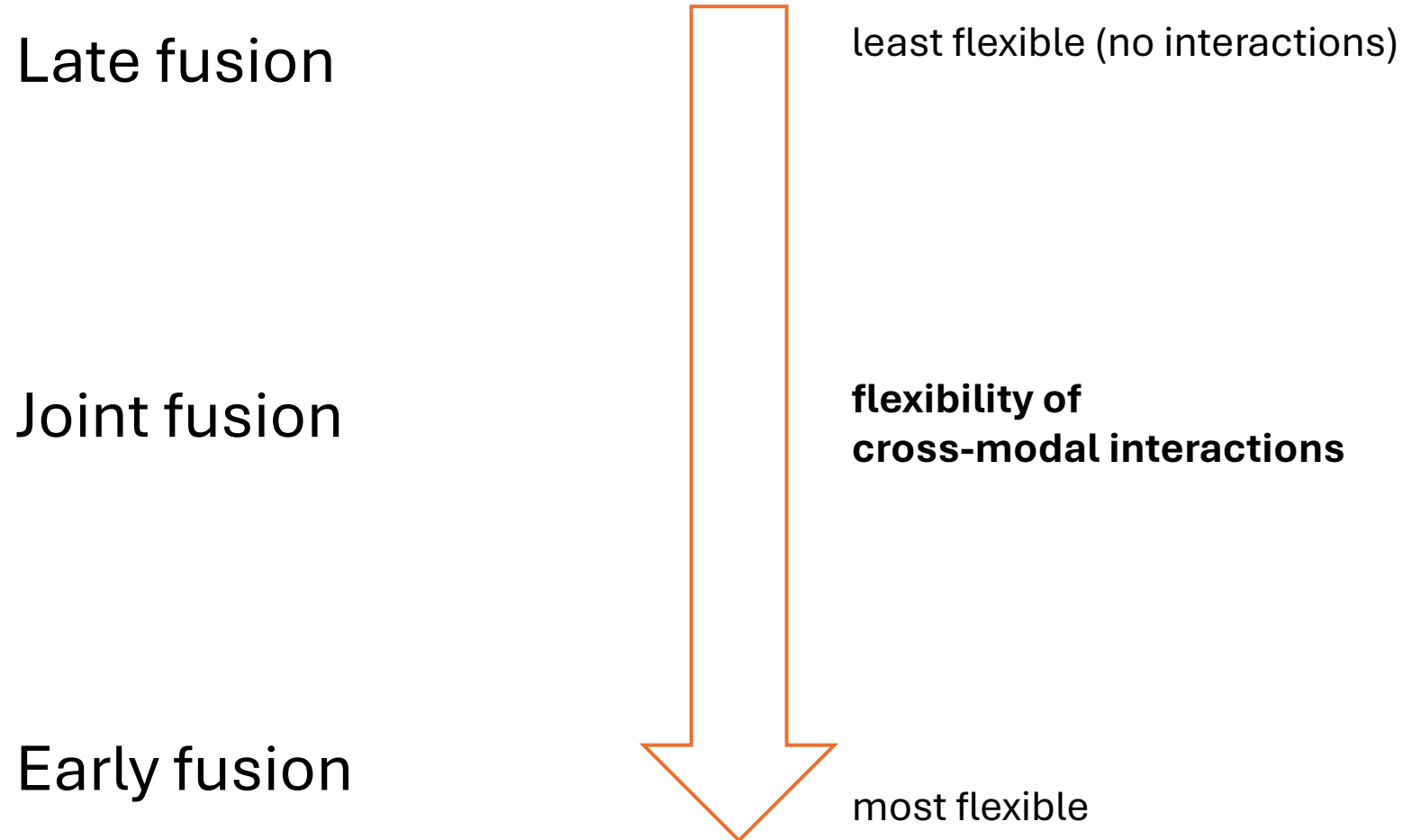
hardest



Comparing fusion methods



Comparing fusion methods



Choosing between fusion strategies

- In most cases, your choice will be between:
 - Late fusion
 - Joint fusion
- Joint fusion requires more data but is usually feasible with health datasets
- Early fusion is likely needed to maximize performance, but it is only realistic with large data and self-supervision

Which fusion method to choose?

- Prediction of retinopathy with *side* information (e.g., demographics)
- Prediction of lupus from information about multiple organ systems (e.g., kidneys, skin, joints)
- Early diagnosis prediction from EHRs
- Cardiovascular event prediction from multiple wearables
- Share examples you have seen in your work!

Discuss Coding Assignment 3

- If needed

Some current directions

(optional content)

Recall: are all tasks special cases of text generation?

ME



Please read the following echocardiogram report again and tell me whether left ventricular hypertrophy was detected. This time, please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that it was NOT detected) to 100% (i.e. you are certain that it WAS detected). Answer with the percentage only.

Echo reading:

MILD LV SYSTOLIC DYSFUNCTION (See above) WITH MILD LVH
NORMAL RIGHT VENTRICULAR SYSTOLIC FUNCTION MODERATE
VALVULAR REGURGITATION (See above) NO VALVULAR STENOSIS
NO PERICARDIAL EFFUSION SEVERE BIATRIAL ENLARGEMENT AT
LEAST MILD PULMONARY HTN ELEVATED RA PRESSURES NO
PRIOR STUDY FOR COMPARISON



100%



Generative LLMs enable “zero-shot” prediction:

i.e. prediction without task-specific training

ME



Please read the following echocardiogram report again and tell me whether left ventricular hypertrophy was detected. This time, please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that it was NOT detected) to 100% (i.e. you are certain that it WAS detected). Answer with the percentage only.

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PRIOR STUDY FOR COMPARISON



100%



Do we still need predictive AI?

and task-specific models?

Often predictive and generative models are complementary.

Maternal health response system:

- Speech to text (predictive)
- Translation (predictive)
- Identification of key concepts and topics (predictive)
- Triage (predictive)
- Generation of template and/or complete responses in specific cases (generative)

It is much easier to evaluate the performance of predictive models, and in turn to know when and how much to trust them.



In some cases, however, generative models have superseded predictive models.

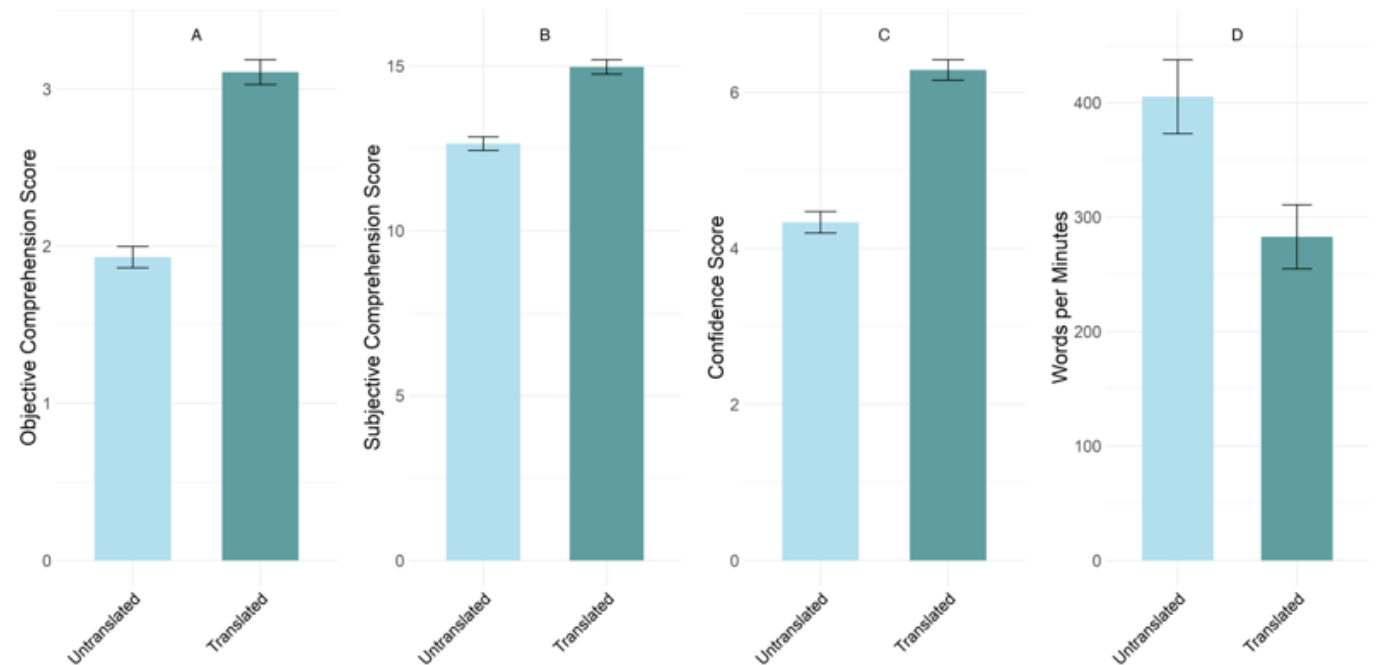
“Translation” of discharge notes into plain language using GPT4 substantially improves patients’ ability to comprehend them.



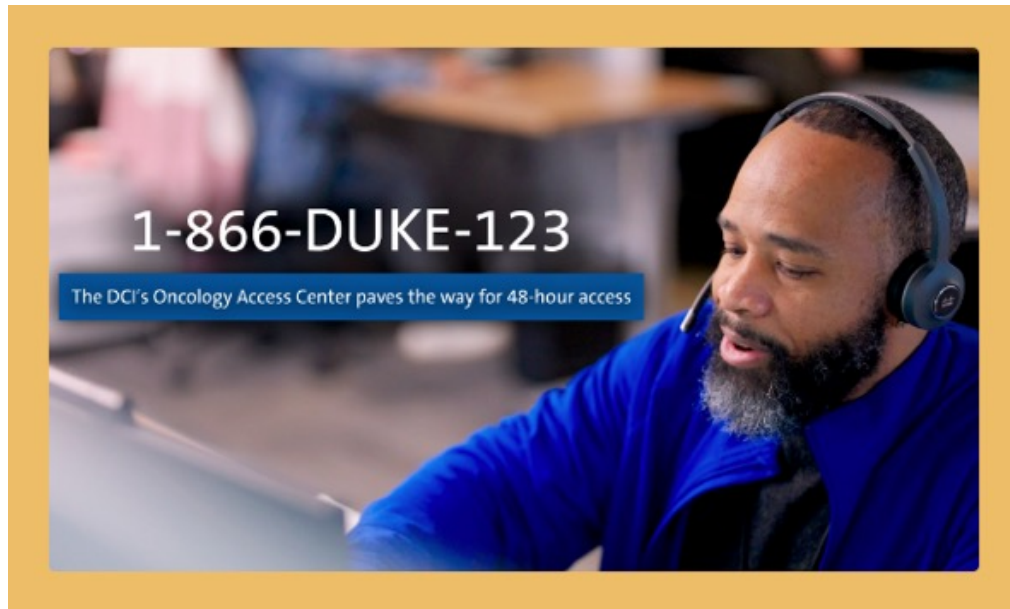
Anivarya Kumar



Isabella Wang



Extracting specific info from documents: predictive or generative?



Data needed to determine scheduling needs
ICD-10
Stage
Age
Date of Dx
Hormone Receptors and molecular biomarkers
History of Treatment (med, surg, rad)
Undiagnosed patients only: Review of symptoms

While a predictive approach might be more trustworthy in principle, generative models can perform many tasks “zero-shot”, i.e. without task-specific training.



DIHI '24, led by Emily Norboge

Current directions: RAG vs long context

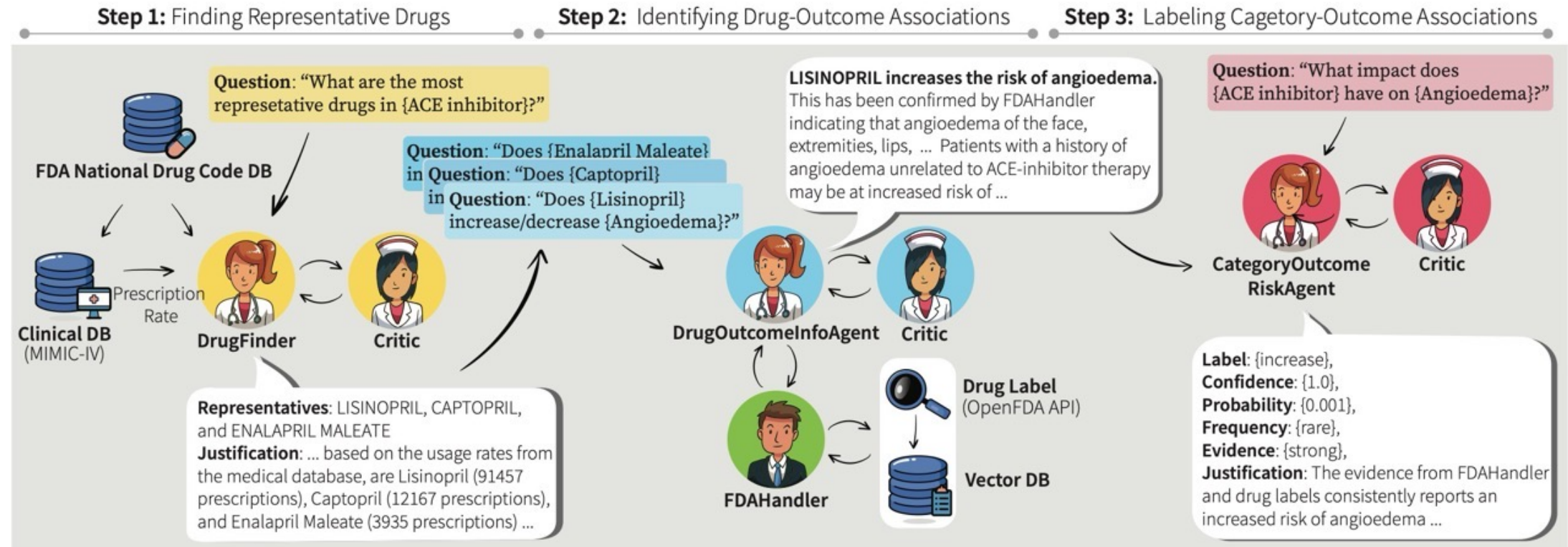
Suppose we wish to answer questions or extract information from a large collection of documents (e.g., all clinical notes for a given patient)

How do we use an LLM to do this?

- Option 1: *very* long context window
- Option 2: retrieval augmented generation (RAG)

Currently, RAG is commonly used because it is effective and scalable, and it (partly) addresses LLM pitfalls, including hallucinations.

Current directions: agentic AI



Multi-LLM-agent systems including:

- An orchestrator agent
- Task-specific agents (e.g. DocChat, SQLchat)
- Critic agents

Current directions: evaluation of generative models

Open questions include:

- Can we trust models to evaluate other models?
- How can models *learn when to defer* to human experts?
- How can we identify LLM biases? (more on this in PIONEER sessions)



Presented by:
Chuan Hong, PhD; Assistant Professor of
Biostatistics & Bioinformatics, Duke
University School of Medicine



AI Health Virtual Seminar Series: Evaluating Generative Large Language Models in Healthcare

Tuesday, April 16, 2024 | 12:00 PM – 1:00 PM (Eastern time)
Virtual seminar via Zoom, open to members internal and external to Duke

Register here: https://duke.zoom.us/webinar/register/WN_4eaOxm6KRXahb8p3H0d3nQ

The rapid evolution of large language models (LLMs) has ushered in a new era of computational linguistics, yet a systematic approach to their evaluation, particularly in sensitive domains such as healthcare, remains nascent. This work bridges these gaps by offering a detailed and integrated review of qualitative evaluation, quantitative evaluation, and meta-evaluation. For quantitative evaluation, our review introduces a taxonomy of evaluation metrics, categorizing them based on essential dimensions such as human supervision, contextual data, and analytical depth. In addition to generic settings, our work distinctively emphasizes additional considerations vital in the healthcare sector. As a result, we propose an integrated cross-walk between qualitative and quantitative assessment methods. The proposed framework harmonizes qualitative insights, such as user-focused evaluations, with objective quantitative metrics. We present a detailed "go-to menu" of evaluation criteria, tailored to address specific healthcare applications and emphasize distinct aspects in both pre-deployment and post-deployment phases. Our findings underscore the need for evaluations that extend beyond mere technical accuracy, factoring in medical ethics, fairness, equity, and potential operational biases. Our work offers a summary of existing methods of LLM evaluation that can establish a baseline from which future evaluation methods can be developed to keep pace with the rapid advancements in the field.

